# COMPETENCY BASED TRAINING CURRICULUM
# FOR
# NVQ LEVEL 5 & 6 NATIONAL DIPLOMA COURSE
# IN
# INFORMATION AND COMMUNICATION TECHNOLOGY

**(CBT Curriculum Code No: K72T001)**

**Developed by**

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**Kandawala Road, Rathmalana**

**Validated by**
**National Apprentice and Industrial Training Authority971, Sri Jayewardenepura Mawatha**
**Welikada**
**Rajagiriya**

**Endorsed by**
**Tertiary and Vocational Education Commission"Nipunatha Piyasa",**
**354/2, Elvitigala Mawatha,**
**Colombo 05.**

**Ministry of Education, Higher Education and Vocational Education.**

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Ministry of Education, Higher Education and Vocational Education

## PREFACE

This curriculum outline is based on National Competency Standards for Information and Communication Technology (CBT Curriculum Code No: K72T001) which specified the skills need of the industry at NVQ Level 5. This is the national curriculum in Competency Based Training (CBT) mode for NVQ Level 5 National Diploma programs under National Vocational Qualification (NVQ) Framework.

The Curriculum outline was developed in consultation with relevant industry experts; it was validated by the National Industrial Training Advisory committee (NITAC) and endorsed by the Tertiary and Vocational Education Commission (TVEC). It is expected to review and revise this document on completion of 3 years. However, all are welcome to point out any shortcomings and propose improvements to this curriculum

In order to implement this curriculum, Training providers is welcome adopting this curriculum with TVEC accreditation and Competency Based assessment. We take this opportunity to thank Minister, Secretary to the Ministry for their policy direction. We appreciate the technical contribution made by the members of the development group and National Industrial Training Advisory Committee and other supporting staff involved in development work. We note with gratitude the hard work done by staff of three institutions to compete this document.

H.A.Saththar
Chairman
National Apprentice and
Industrial Training Authority

Dr. K.A. Lalithadeera
Director General
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# ACKNOWLEDGMENT


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| **Reviewing and edited by** | • Mrs. A.H.Kaushika<br>Training Officer (SSCD), National Apprentice and Industrial Training Authority, Rajagiriya |

# Competency Based Training Curriculum
# for
# NVQ Levels 5 & 6 National Diplomas in Information and Communication Technology
# Implementation of NVQ Level 5 & 6 Curricula

Training providers are required to meet following conditions in order to implement NVQ level5 & 6 diplomas;

i. Credit Policy

NVQ Level 5 and 6 programs should have minimum 60 credits (1500 Notional Hours) and 120 minimum credits (3000 notional hours) respectively as details given below.

| **Levels of Credits** | **NVQ Level 5 National Diploma** | **NVQ Level 6 National Diploma** |
|---|---|---|
| Level 3 & 4 credit (Maximum) | 20 Credits (500 hrs.) | 20 Credits (500 hrs.) |
| Level 5 credit (Minimum) | 40 Credits (1000 hrs.) | 50 Credits (1250 hrs.) |
| Level 6 credit (Minimum) | - | 50 Credits (1250 hrs.) |
| Total Number of credit (Minimum) | 60 Credits (1500 hrs.) | 120 Credits (3000 hrs.) |

Note: one credit is equivalent to 25 notional hours which include class room teaching, tutorials, assignment, projects, on the job training and self-study.

Credit policy defines the minimum course content. Training providers could add any more subjects above modular requirements of the qualification packages.

ii. Units Classifications

List of Units consists of Core Technical Units, Electives and Employability Units.

Core Technical Units in NCS : It is compulsory to include these units in Qualification

Employability Unit : It is compulsory to include 8 Credit in NVQ 5 and 6 credit in NVQ 6. These units are listed in the NCS.

Electives : Units to be added to make total credits equivalent to 60, if total credits of Core Technical Units and Employability units are less than 60.

iii. Approval to establish NVQ Level 5 & 6 Course

Training providers are required submit a proposal based on Guideline 01 – "Criteria on Requirements to conduct NVQ Level 5 & 6 Programs" which could be downloaded from TVEC website and obtain approval from TVEC to establish the course.

iv. Entry Qualification

Relevant NVQ Level 4 Qualification or 3 passes at GCE (A/L). Details of relevant NVQ 4 and subjects of GCE (A/L) relevant to this course could be obtained from TVEC. Bridging modules should be organized as required in the pre-requisites in modules.

v. Upon establishment of the course, obtain TVEC Accreditation and Install a Quality Management System

vi. Conduct continuous assessment and maintain a progress record book

vii. Written test at the end of each semester conducted by the training institute, with paper moderation by TVEC. Pass Mark for written tests is 50%

viii. Three Months on the job training on completion of both NVQ Level 5 and NVQ Level 6 programs respectively.

ix. Competency Based Assessment (Final Assessment) by NVQ Level 5 & 6 Assessors assigned by TVEC.

x. Certificates are issued by TVEC with the signature of the Director General of TVEC and Head and Officer responsible for assessment at the training centre.

## Competency Based Training Curriculum
## for
## NVQ Levels 5 National Diploma in Information and Communication Technology


| Technical Modules | | | |
|---|---|---|---|
| No | List of Modules | Code No. | Page |
| 1. | User requirements for software solution | K72T001M01 | 01 |
| 2. | System analysis and design | K72T001M02 | 03 |
| 3. | Mathematics For Computing | K72T001M03 | 05 |
| 4. | Digital Electronics and Computer Architecture | K72T001M04 | 07 |
| 5. | Object Oriented Programming Concepts | K72T001M05 | 09 |
| 6. | Data Structures and Algorithms | K72T001M06 | 11 |
| 7. | Web Site Design and Hosting | K72T001M07 | 14 |
| 8. | Creative design for print media | K72T001M08 | 17 |
| 9. | Software Testing & Quality Assurance | K72T001M09 | 22 |
| 10. | Web Application Development | K72T001M10 | 24 |
| 11. | Database Management Systems | K72T001M11 | 27 |
| 12. | Multimedia production | K72T001M12 | 30 |
| 13. | Computer Networks | K72T001M13 | 34 |
| 14. | Cyber Security and Cloud computing | K72T001M14 | 37 |
| 15. | Developing Mobile App | K72T001M15 | 39 |
| Employability Modules in Level 5 | | | |
| 16. | Workplace information management | EMPM01 | 42 |
| 17. | Workplace communication management | EMPM02 | 20 |
| 18. | Planning and scheduling work at workplace | EMPM03 | 45 |

**Competency Based Training Curriculum**
**for**
**NVQ Levels 6 National Diploma in Information and Communication Technology**


| Technical Modules | | | |
|---|---|---|---|
| No | List of Modules | Code No. | Page |
| 1. | IoT Development | K72T001M16 | 48 |
| 2. | Project Management | K72T001M17 | 50 |
| 3. | Relational and NoSQL Databases | K72T001M18 | 53 |
| 4. | Information Security concepts and Ethics | K72T001M19 | 55 |
| 5. | Manage Secured Mobile App Development | K72T001M20 | 58 |
| 6. | Green IT Concept | K72T001M21 | 60 |
| 7. | Advanced Techniques in Graphics Designing for Multimedia Content | K72T001M22 | 63 |
| 8. | Design and develop audio, video, motion graphics and amination for multimedia content | K72T001M23 | 66 |
| 9. | Manage cloud resources | K72T001M24 | 69 |
| 10. | Enterprise network setup and configuration | K72T001M25 | 72 |
| 11. | Maintain Network solution | K72T001M26 | 74 |
| 12. | Operating Systems and Application Security | K72T001M27 | 77 |
| 13. | DevOps and Site Reliability Engineering (SRE) | K72T001M28 | 83 |
| 14. | Software Development final Project | K72T001M29 | 86 |
| 15. | Artificial Intelligence (AI) | K72T001M30 | 88 |
| 16. | Business Continuity Planning and Disaster Recovery | K72T001M31 | 90 |
| 17. | Generate Analytical Reports Using Data Science Tools | K72T001M32 | 93 |
| 18. | Design, Develop and Maintain Web Application | K72T001M33 | 95 |
| 19. | Advertising and Campaign Management | K72T001M34 | 98 |
| 20. | Analyze and finalize the cost estimate of the Network solution | K72T001M35 | 101 |
| 21. | Fundamentals of Information, Communication & Network Security (Governance Risk and Compliance) | K72T001M36 | 104 |
| 22. | Ethical Hacking, Mobile and Internet of Things (IoT) Security | K72T001M37 | 107 |
| Employability Modules in Level 6 | | | |
| 23. | Problem Solving and Decision Making | EMPM04 | 80 |
| 24. | Teamwork and leadership | EMPM05 | 109 |
| 25. | Creation and maintenance of a learning culture | EMPM06 | 112 |

# MODULE STRUCTURE
# NVQ LEVEL 5 QUALIFICATION

**Semester 1**

| Module Code | Module | Type C/E | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M01 | User requirements for software solution | C | 100 | 60 | 30 | 10 |
| K72T001M02 | System analysis and design | C | 100 | 60 | 30 | 10 |
| K72T001M03 | Mathematics For Computing | C | 25 | 20 | 00 | 05 |
| K72T001M04 | Digital Electronics and Computer Architecture | C | 75 | 42 | 18 | 15 |
| K72T001M05 | Object Oriented Programming Concepts | C | 100 | 54 | 36 | 10 |
| K72T001M06 | Data Structures and Algorithms | C | 100 | 60 | 30 | 10 |
| K72T001M07 | Web Site Design and Hosting | C | 100 | 60 | 30 | 10 |
| K72T001M08 | Creative design for print media | | 100 | 30 | 60 | 10 |
| EMPM02 | Workplace Communications Management | C | 50 | 30 | 15 | 05 |
| Total | | | 750 | 416 | 249 | 85 |

**Semester 2**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M09 | Software Testing & Quality Assurance | C | 75 | 42 | 18 | 15 |
| K72T001M10 | Web Application Development | C | 100 | 60 | 30 | 10 |
| K72T001M11 | Database Management Systems | C | 100 | 60 | 30 | 10 |
| K72T001M12 | Multimedia production | C | 100 | 30 | 60 | 10 |
| K72T001M13 | Computer Networks | C | 100 | 60 | 30 | 10 |
| K72T001M14 | Cyber Security and Cloud computing | C | 50 | 24 | 18 | 08 |
| K72T001M15 | Developing Mobile App | C | 75 | 50 | 20 | 05 |
| EMPM01 | Workplace information management | C | 75 | 45 | 20 | 10 |
| EMPM03 | Planning and Scheduling Work at Workplace | C | 75 | 45 | 20 | 10 |
| Total | | | **750** | **416** | **246** | **88** |

C – Compulsory

# MODULE STRUCTURE
# NVQ LEVEL 6 QUALIFICATION

**Package 01: Software Development with Data Science**

**Semester 1**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M16 | IoT Development | C | 150 | 60 | 60 | 30 |
| K72T001M17 | Project Management | C | 100 | 60 | 30 | 10 |
| K72T001M18 | Relational and NoSQL Databases | C | 150 | 84 | 36 | 30 |
| K72T001M19 | Information Security concepts and Ethics | C | 100 | 48 | 30 | 22 |
| K72T001M20 | Manage Secured Mobile App Development | C | 100 | 48 | 48 | 04 |
| K72T001M21 | Green IT Concept | C | 100 | 48 | 30 | 22 |
| EMPM04 | Problem Solving and Decision Making | C | 50 | 30 | 15 | 05 |
| **Total** | | | **750** | **378** | **249** | **123** |

**Semester 2**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M28 | DevOps and Site Reliability Engineering (SRE) | C | 100 | 72 | 20 | 08 |
| K72T001M29 | Software Development final Project | C | 200 | 50 | 00 | 150 |
| K72T001M30 | Artificial Intelligence (AI) | C | 150 | 75 | 60 | 15 |
| K72T001M31 | Business Continuity Planning and Disaster Recovery | C | 100 | 60 | 25 | 15 |
| K72T001M32 | Generate Analytical Reports Using Data Science Tools | C | 100 | 54 | 34 | 12 |
| EMPM05 | Teamwork and Leadership | C | 50 | 30 | 15 | 05 |
| EMPM06 | Creation and maintenance of learning culture | C | 50 | 30 | 15 | 05 |
| **Total** | | | **750** | **371** | **169** | **210** |

C- Compulsory

# MODULE STRUCTURE
# NVQ LEVEL 6 QUALIFICATION

**Package 02: Web Design and Development**

**Semester 1**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M17 | Project Management | C | 100 | 60 | 30 | 10 |
| K72T001M22 | Advanced Techniques in Graphics Designing for Multimedia | C | 200 | 90 | 80 | 30 |
| K72T001M23 | Design and develop audio, video, motion graphics and amination for multimedia content | C | 200 | 90 | 80 | 30 |
| K72T001M24 | Manage cloud resources | C | 100 | 60 | 30 | 10 |
| K72T001M21 | Green IT Concept | C | 100 | 48 | 30 | 22 |
| EMPM04 | Problem Solving and Decision Making | C | 50 | 30 | 15 | 05 |
| **Total** | | | **750** | **378** | **265** | **107** |

**Semester 2**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M33 | Design, Develop and Maintain Web Application | C | 175 | 90 | 75 | 10 |
| K72T001M34 | Advertising and Campaign Management | C | 125 | 75 | 40 | 10 |
| K72T001M24 | Fundamentals of Artificial Intelligence (AI) | C | 150 | 75 | 60 | 15 |
| K72T001M25 | Business Continuity Planning and Disaster Recovery | C | 100 | 60 | 25 | 15 |
| K72T001M26 | Generate Analytical Reports Using Data Science Tools | C | 100 | 54 | 34 | 12 |
| EMPM05 | Teamwork and Leadership | C | 50 | 30 | 15 | 05 |
| EMPM06 | Creation and maintenance of learning culture | C | 50 | 30 | 15 | 05 |
| **Total** | | | **750** | **414** | **264** | **72** |

C- Compulsory

## MODULE STRUCTURE
## NVQ LEVEL 6 QUALIFICATION

**Package 03: Networking, Cloud computing & Cyber Security**

**Semester 1**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M17 | Project Management | C | 100 | 60 | 30 | 10 |
| K72T001M25 | Enterprise network setup and configuration | C | 100 | 66 | 22 | 12 |
| K72T001M26 | Maintain Network solution | C | 100 | 60 | 30 | 10 |
| K72T001M24 | Manage cloud resources | C | 100 | 60 | 30 | 10 |
| K72T001M27 | Operating Systems and Application Security | C | 100 | 60 | 30 | 10 |
| K72T001M19 | Information Security concepts and Ethics | C | 100 | 48 | 30 | 22 |
| K72T001M21 | Green IT Concept | C | 100 | 48 | 30 | 22 |
| EMPM04 | Problem Solving and Decision Making | C | 50 | 30 | 15 | 05 |
| **Total** | | | **750** | **432** | **217** | **101** |

**Semester 2**

| Module Code | Module | Type | Notional hours | Lectures/ Tutorials | Practical/Designs/ Industry visits | Self-study |
|---|---|---|---|---|---|---|
| K72T001M24 | Fundamentals of Artificial Intelligence (AI) | C | 150 | 75 | 60 | 15 |
| K72T001M35 | Analyze and finalize the cost estimate of the Network so | C | 100 | 57 | 30 | 13 |
| K72T001M36 | Fundamentals of Information, Communication & Network Security (Governance Risk and Compliance) | C | 100 | 60 | 30 | 10 |
| K72T001M37 | Ethical Hacking, Mobile and Internet of Things (IoT) Security | C | 100 | 60 | 30 | 10 |
| K72T001M25 | Business Continuity Planning and Disaster Recovery | C | 100 | 60 | 25 | 15 |
| K72T001M26 | Generate Analytical Reports Using Data Science Tools | C | 100 | 54 | 34 | 12 |
| EMPM05 | Teamwork and Leadership | C | 50 | 30 | 15 | 05 |
| EMPM06 | Creation and maintenance of learning culture | C | 50 | 30 | 15 | 05 |
| **Total** | | | **750** | **426** | **239** | **85** |

C- Compulsory

## MAPPING OF COMPETENCY UNITS AND CURRICULUM MODULES
## NVQ LEVEL 5 QUALIFICATION

**Semester 1**

| Module Code | Module | Related Competency Unit/s |
|---|---|---|
| K72T001M01 | User requirements for software solution | K72T001U01 |
| K72T001M02 | System analysis and design | K72T001U02 |
| K72T001M03 | Mathematics For Computing | All Units |
| K72T001M04 | Digital Electronics and Computer Architecture | All Units |
| K72T001M05 | Object Oriented Programming Concepts | K72T001U04, K72T001U07 |
| K72T001M06 | Data Structures and Algorithms | K72T001U04, K72T001U07 |
| K72T001M07 | Web Site Design and Hosting | K72T001U06, K72T001U07 |
| K72T001M08 | Creative design for print media | K72T001U05 |
| EMPM02 | Workplace Communications Management | EMPU02 |

**Semester 2**

| Module Code | Module | Related Competency Unit/s |
|---|---|---|
| K72T001M09 | Software Testing & Quality Assurance | K72T001U04, K72T001U07 |
| K72T001M10 | Web Application Development | K72T001U07 |
| K72T001M11 | Database Management Systems | K72T001U03 |
| K72T001M12 | Multimedia production | K72T001U06, K72T001U07 |
| K72T0011M13 | Computer Networks | K72T001U08 |
| K72T001M14 | Cyber Security and Cloud computing | K72T001U08 |
| K72T001M15 | Developing Mobile App | K72T001U09 |
| EMPM01 | Workplace Information Management | EMPU01 |
| EMPM03 | Planning and scheduling work at workplace | EMPU03 |

## MAPPING OF COMPETENCY UNITS & CURRICULUM MODULES
## NVQ LEVEL 6 QUALIFICATION

**Semester 1**

| Module Code | Module | Related Competency Unit/s |
|---|---|---|
| K72T001M16 | IoT Development | K72T001U15 |
| K72T001M17 | Project Management | K72T001U14 |
| K72T001M18 | Relational and NoSQL Databases | K72T001U11 |
| K72T001M19 | Information Security concepts and Ethics | K72T001U24, K72T001U25, K72T001U26, K72T001U27 |
| K72T001M20 | Manage Secured Mobile App Development | K72T001U13 |
| K72T001M21 | Green IT Concept | All Units |
| K72T001M22 | Advanced Techniques in Graphics Designing for Multimedia | K72T001U17, K72T001U18, K72T001U19 |
| K72T001M23 | Design and develop audio, video, motion graphics and amination for multimedia content | K72T001U16 |
| K72T001M24 | Manage cloud resources | K72T001U23 |
| K72T001M25 | Enterprise network setup and configuration | K72T001U21 |
| K72T001M26 | Maintain Network solution | K72T001U22 |
| K72T001M27 | Operating Systems and Application Security | K72T001U25 |
| EMPM04 | Problem Solving and Decision Making | EMPU04 |

**Semester 2**

| Module Code | Module | Related Competency Unit/s |
|---|---|---|
| K72T001M28 | DevOps and Site Reliability Engineering (SRE) | K72T001U12 |
| K72T001M29 | Software Development final Project | K72T001U10, K72T001U11, K72T001U12, K72T001U13, K72T001U14, K72T001U15 |
| K72T001M30 | Artificial Intelligence (AI) | All Units |
| K72T001M31 | Business Continuity Planning and Disaster Recovery | All Units |
| K72T001M32 | Generate Analytical Reports Using Data Science Tools | K72T001U12 |
| K72T001M33 | Design, Develop and Maintain Web Application | K72T001U17, K72T001U18 |
| K72T001M34 | Advertising and Campaign Management | K72T001U16, K72T001U17, K72T001U18, K72T001U19 |
| K72T001M35 | Analyze and finalize the cost estimate of the Network solution | K72T001U20 |
| K72T001M36 | Fundamentals of Information, Communication & Network Security (Governance Risk and Compliance) | K72T001U24 |
| K72T001M37 | Ethical Hacking, Mobile and Internet of Things (IoT) Security | K72T001U25, K72T001U27 |
| EMPM05 | Teamwork and leadership | EMPU05 |
| EMPM06 | Creation & maintenance of a learning culture | EMPU06 |

# MODULE DESCRIPTORS
# Information and Communication Technology
# NVQ Level 05
## Semester I
## Technical modules

| | |
|---|---|
| **Module Title** | **User requirements for software solution** |
| **Module Code** | K72T001M01/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U01 |
| **Pre-Requisites** | Relevant NVQ Level 4 Qualification or GCE (A/L) or equivalent qualification |
| **Module Aim/s** | **To enable the students to:**<br>Identify business processes of various organizations, gather and analyze requirement of selected business organizations and conduct feasibility study. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain the business processes of a selected domain<br>2. Prepare and present requirement analysis report of the selected project<br>3. Prepare and present feasibility study report of the selected project |
| **Learning Content / Topics** | • Introduction to various business processes - (15hrs.)<br>o Business domains (E.g. Telecommunication, Hotels, Health, Transport, Super markets, Education, etc.)<br>o Involvement of the stakeholders to the business process<br>o Software solutions to improve the business functions<br>• Study of business processes –(15hrs.)<br>o Study current system /processes<br>o Roles of Stakeholders<br>o Study business process re-engineering options<br>• Requirements Gathering - (12hrs.)<br>o Importance of gathering requirement<br>o Requirements gathering techniques (Document review, Interviews, Surveys and Questionnaires, Observation, etc.)<br>• Requirement analysis - (12hrs.)<br>o Project Scope (Objectives, Specifications, Boundaries, Limitations)<br>o Functional and non-functional requirements<br>• Feasibility study - (6hrs.)<br>o Technical, Economical, Practical, Organizational, Legal, etc. |
| **Practical** | • Assignment:<br>o Assignment 01 - Prepare a user requirement document<br>o Assignment 02 - Prepare a feasibility study report for a given industry specific scenario |
| **Resources** | • Multimedia, Whiteboard<br>• Smart board<br>• Computers |

<table>
<tr><td></td><td colspan="3">• Computer hardware components</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Software Requirements Essentials, Karl Wiegers, Candase Hokanson, 2023, ISBN-13: 978-0-13-819028-6 / ISBN-10: 0-13-819028-3</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">• Interactive Lecture with hands on practical sessions<br>• Demonstration/practical<br>• Individual assignment<br>• Simulation<br>• Programmed instructions</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Assignment 01</td><td>30%</td></tr>
<tr><td>Assignment 02</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam– Minimum 2hour</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T + 30 P + 10 S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **System analysis and design** |
| **Module Code** | K72T001M02/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U02 |
| **Pre-Requisites** | K72T001M01 |
| **Module Aim/s** | **To enable the students to:**<br>Apply concepts related to identify, analyze, prioritize, document, and manage user requirements for software solutions by applying requirement gathering and requirement identification techniques. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>• Analyze requirements specified in documentation<br>• Apply system analysis techniques for logical design<br>• Identify components and documenting functionality and data<br>• Document requirement outcomes according to standard templates<br>• Develop test cases based on design specifications<br>• Prepare the final requirement document |
| **Learning Content / Topics** | • Introduction to systems Analysis -(20hrs.)<br>o System analysis techniques<br>▪ DFD<br>▪ Use case diagrams<br>▪ ER/EER<br>o Introduction to systems design –(20hrs.)<br>▪ Flow chart and pseudocodes<br>▪ UI/UX Design<br>o System requirements specification document-(20hrs.)<br>▪ Executive summery<br>▪ Summary of user requirements<br>▪ Proposed solution<br>▪ Schedule structures |
| **Practical** | Group Assignment: Prepare SRS document for a given user requirement document (Can use the user requirement document prepared in K72T003M01) |
| **Resources** | • Multimedia, Whiteboard<br>• Smart board<br>• Computers<br>• Computer hardware components |
| **Prescribed Texts & / or References** | • Modern systems analysis and design, Joseph S. Valacich,10th Edition, ISBN 9780138179946<br>• Systems Analysis and Design, Scott Tilley, Twelfth Edition, ISBN: 978-0-357- |

<table>
<tr><td></td><td colspan="3">11781-1<br>• Systems Analysis and Design, Scott Tilley and Harry Rosenblatt, Eleventh Edition, ISBN: 978-1-305-49460-2<br>• Systems Analysis & Design an Object-Oriented Approach with UML, Alan Dennis, Sixth Edition, ISBN: 978-1-119-55991-7</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">• Interactive Lecture with hands on practical sessions<br>• Demonstration/practical<br>• Individual assignment<br>• Simulation<br>• Programmed instructions</td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/</td><td>Module Weighting</td></tr>
<tr><td>Continuous Assessment</td><td>Group Assignment</td><td>60%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (Minimum 2hour Paper)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T + 30P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Mathematics for Computing** |
| **Module Code** | K72T001M03/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | All Units |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>• Understand the different types of number systems<br>• Understand various computer coding systems<br>• Understand basic statistics concepts |
| **Learning Content / Topics** | • Types of Number Systems -(01Hrs.)<br>  ○ Positional and non-positional<br>  ○ Factors affecting to the value of the number<br>• Conversion between different bases -(06Hrs.)<br>  ○ Decimal to other bases (binary, octal, hexadecimal)<br>  ○ Other bases to decimal<br>  ○ Non-decimal bases to non-decimal bases<br>  ○ Fractional numbers<br>• Calculation using different number bases -(04Hrs.)<br>  ○ Adding numbers (binary, octal, hexadecimal)<br>  ○ Subtracting numbers<br>  - One's complement<br>  - Two's complement<br>• Fractional number representation -(02Hrs.)<br>  ○ Fixed point and floating point<br>• Sets theory -(02Hrs.)<br>• Statistics fundamentals -(03Hrs.)<br>  ○ Mean, Median, Mode, Standard Deviation<br>• Computer Codes -(02Hrs.)<br>  ○ BCD, EBCDIC, ASCII, Unicode |
| **Practical** | None |

<table>
<tr><td>Resources</td><td colspan="3"><ul><li>Text books</li><li>Model Papers</li></ul></td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>Discrete Mathematics for Computer Science, An Example-Based Introduction, Jon Pierre Fortney, ISBN: 978-1-003-09147-9</li></ul></td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3"><ul><li>Lectures</li><li>Presentations</li><li>Discussions</li></ul></td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td>Formative Assessment</td><td>Assignment</td><td>40%</td></tr>
<tr><td>Summative Assessment (Semester end)</td><td>Written test (2 hours)</td><td>60%</td></tr>
<tr><td>Duration</td><td colspan="3">25hrs. (20T + 0P + 05S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Digital Electronics and Computer Architecture** |
| **Module Code** | K72T001M04/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | All Units |
| **Pre-Requisites** | K72T001M03 |
| **Module Aim/s** | **To enable the students to:**<br>• Understand the logic gates<br>• Describe basic computer component and functional organization<br>• Describe processor organization<br>• Familiarize with software systems and IoT integration |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Demonstrate an understanding of logic gates<br>2. Design a logic circuit for a given scenario |
| **Learning Content / Topics** | • Logic Gates –(12hrs)<br>o Basic Logic Gates<br>o Universal Logic Gates<br>o Simplification of Boolean Expressions<br>➢ Boolean Algebra laws<br>➢ Karnaugh maps<br>o Sequential logic circuits<br>o Design logic circuits using NAND / NOR gates<br>• Computer Organization and Architecture –(21hrs)<br>o Processor organization<br>o Fetch and Execution Cycle and program execution<br>o Instructions Set Architecture<br>o Memory organization<br>o Input output organization<br>o Performance evaluation of program execution<br>o Pipeline architecture<br>o High performance computers and multicore architectures<br>• Von-Neumann architecture –(03hrs)<br>• Beyond Von-Neumann architecture –(06hrs)<br>o Quantum computers<br>o Neural Networks<br>o Nature inspired computing |
| **Practical** | Assignment - Design a logic circuit using relevant software for a given Boolean expression |
| **Resources** | • Computer |

<table>
<tr><td></td><td colspan="3"><ul><li>Internet Facility</li><li>Relevant Software</li></ul></td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>Computer Organization and Architecture Designing for Performance, William Stallings, Eleventh Edition, ISBN 13: 978-1-292-42008-0</li><li>Essentials of Computer Architecture, Douglas Comer, Third Edition, ISBN: 978-1-0034-1014-0</li><li>The Essentials of Computer Organization and Architecture Linda Null, 6th Edition, ISBN 9781284259438</li></ul></td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3"><ul><li>Lectures</li><li>Presentations</li><li>Demonstration</li><li>Discussions using the materials/ components and product samples</li></ul></td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td>Continuous Assessment</td><td>Assignment</td><td>50%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam</td><td>50%</td></tr>
<tr><td>Duration</td><td colspan="3">75hrs. (42T + 18P + 15S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Object Oriented Programming Concepts** |
| **Module Code** | K72T001M05/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | All Units |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Understand the basic program structures (Sequence, Selection, Iteration)<br>• Understand the syntax of selected programming language |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>• Draw flowcharts and write pseudocodes<br>• Write relevant program codes<br>• Test the developed program/s<br>• Apply IDEs for program development |
| **Learning Content / Topics** | • Basics of computer programming – (18hrs)<br>o Algorithms<br>o Flowcharts, Pseudocodes<br>• Introduction to Object-Oriented Programming – (06hrs)<br>o Comparison between procedural and object-oriented paradigms<br>• OOP concepts – (24hrs)<br>o Classes and Objects<br>- Real-world examples of classes and objects<br>- Variables and Data Types<br>- Attributes and Methods<br>- Instantiating objects from classes<br>- Accessing attributes and methods<br>- Constructors and Destructors<br>- Overloading<br>o Memory Management in OOP<br>- Dynamic memory allocation<br>- Garbage collection and object lifecycle<br>o Modifiers<br>- Access Modifiers<br>- Non-access modifiers<br>o Encapsulation<br>- Getters and setters<br>o Inheritance<br>- Hierarchies and class relationships<br>- Single, Multiple inheritance<br>- Overriding methods<br>o Polymorphism |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Abstraction</li>
<li>Abstract Classes and Interfaces</li>
<li>Real-world Examples of Encapsulation and Abstraction</li>
</ul>
• Association, aggregation and composition – (06hrs)
</td></tr>
<tr><td>Practical</td><td colspan="3">Assignment - Write codes to implement for the designed system</td></tr>
<tr><td>Resources</td><td colspan="3">
• Computer<br>
• Internet Facility<br>
• Relevant Software<br>
• Text Books / Reference Books<br>
• Question Papers
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
• Hands-On Object-Oriented Programming, Anil Kumar Rangisetti, ISBN-13: 979-8-8688-0524-0<br>
• Python 3 Object-Oriented Programming, Dusty Phillips, Third Edition, ISBN 978-1-78961-585-2<br>
• Java Essentials For Dummies, Doug Lowe and Paul McFedries, ISBN: 978-1-394-29699-6<br>
• Java Foundations, Fifth Edition, John Lewis, ISBN-13: 9780137518401<br>
• Getting Started with Python: Understand key data structures and use Python in object-oriented programming, Romano, Fabrizio;Baka, Benjamin; Phillips, Dusty, ISBN: 9781838551919<br>
• Modern Python Cookbook, Steven F. Lott, Second Edition, 9781800207455
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
• Lectures<br>
• Presentations<br>
• Discussions using the material and questions samples
</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Formative Assessment</td><td>Assignment</td><td>30%</td></tr>
<tr><td>Practical test</td><td>40%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam</td><td>30%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (54T + 36P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Data Structures and Algorithms** |
| **Module Code** | K72T001M06/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U04, K72T001U07 |
| **Pre-Requisites** | None |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Apply fundamental programming concepts<br>2. Draw flowcharts to sort a set of numbers / names<br>3. Draw flowcharts to find a number / name<br>4. Write pseudocodes to sort a set of numbers / names<br>5. Write pseudocodes to find a number / name |
| **Learning Content / Topics** | • Introduction to Software Development – (06hrs.)<br>  ○ What is software development?<br>  ○ Software Development Life Cycle (SDLC) models<br>  ▪ Waterfall, Agile, Scrum, etc.<br>• Basics of Programming - (06hrs.)<br>  ○ Introduction to programming languages<br>  ▪ Structured/Object-Oriented programming languages<br>  ○ Setting up development environments (IDE, code editors)<br>  ○ Writing and executing basic programs<br>  ○ Flowcharts<br>  ○ Pseudocode<br>• Data structures (06hrs)<br>  ○ Records<br>  ○ Arrays<br>  ○ Queues & Stacks<br>  ○ Link List<br>  ○ Trees<br>  ○ Maps<br>  ○ Tuples<br>  ○ Sets<br>• Sorting and searching algorithms (06hrs)<br>  ○ Bubble<br>  ○ Selection<br>  ○ Insertions<br>  ○ Quick sort<br>  ○ Hashing Techniques<br>  ○ Linear search<br>  ○ Binary search<br>• Programming Concepts (Use Python, Java, PHP) - (18hrs.)<br>  ○ Variables, data types<br>  ○ Operators (Mathematical, Comparison and Logical) |

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| | ○ Control flow statements<br>▪ Conditions –if..then..else, case/switch /match<br>▪ Nested Conditions<br>▪ Loops –for, while, until<br>▪ Nested Loops<br>○ Functions and Modular Programming<br>▪ Defining and invoking functions<br>▪ Parametrized functions<br>▪ Scope and lifetime of variables<br>▪ Recursion<br>• Introduction to Exception Handling - (06hrs.)<br>○ Basic concepts of exceptions and error handling<br>• Connecting the application to the database - (06hrs.)<br>• Version Control Systems - (06hrs.)<br>○ Introduction to Git/Git hub |
| **Practical** | ○ Write a program to sort a set of numbers or names<br>○ Write a program to find a given number or name in a data set<br>○ Write a program for a full functional calculator |
| **Resources** | • Multimedia, Whiteboard<br>• Smart board<br>• Computers with relevant software |
| **Prescribed Texts & / or References** | • Introduction to Algorithms, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Fourth Edition, ISBN 9780262046305<br>• Absolute Beginner's Guide to Algorithm, Kirupa Chinnathambi, ISBN: 978-0-13-8222229-1<br>• Data Structures and Algorithms Made Easy with Java: Learn Data Structure using Java in 7 Days: Data Structures and Algorithmic Puzzles for Beginners to Professional, Maurya, Rahul, (No ISBN)<br>• Advanced Data Structures and Algorithms, Abirami A, Priya R L, ISBN: 978-93-5551-792-0<br>• The Algorithm Design Manual, Steven S. Skiena, Second Edition, ISBN: 978-1-84800-069-8 |
| **Teaching/ Learning Activities** | Practical:<br>• Practical 01- Interactive Lecture with hands on practical sessions<br>• Practical 02 -Demonstration/practical<br>• Practical 03 - Individual/Group assignment<br><br>Assignment:<br>Assignment 01- Draw a flow chart to sort a set of numbers or names using a standard sorting algorithm<br>Assignment 02-Draw a flow chart to find the 3rd largest number in a data set |

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| **Assessment & Weighting** | **Type** | **Topic / Activity/** | **Module Weighting** |
| | Continuous Assessment | Assignment 01 | 10% |
| | | Assignment 02 | 10% |
| | | Mini project (Individual)- Develop a software application with database connection for a given scenario (Use Python/ Java/ PHP) | 40% |
| | Summative Assessment | Written Test (3 Hour) | 40% |
| **Duration** | **100hrs. (60T + 30 P + 10 S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

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|---|---|
| **Module Title** | **Web Site Design and Hosting** |
| **Module Code** | K72T001M07/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U06, K72T001U07 |
| **Pre-Requisites** | None |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Design a basic web page/site using HTML<br>2. Stylize the web pages applying relevant CSS<br>3. Design a basic web page/site using a web authoring tool/CMS<br>4. Host the developed web site on a web host |
| **Learning Content / Topics** | • Introduction to Web Design - (03Hrs.)<br>o Importance of the Web Design<br>o Overview of Web Technologies<br>o Understanding User Experience (UX) and User Interface (UI)<br>o Tools for Web Design<br>▪ Web authoring tools, Content Management Systems (CMS)<br>o Web design trends and best practices<br><br>• Wireframing and Prototyping - (06Hrs.)<br>o Introduction to Wireframing and Prototyping<br>o Tools use for Wireframing and Prototyping<br>o Draw wireframes<br>o Develop Prototypes<br><br>• Fundamentals of Web Technologies- (03Hrs.)<br>o The World Wide Web (WWW) and its components (browsers, servers, URLs)<br>o Client-side vs. server-side scripting<br><br>• Hypertext Markup Language (HTML) - (15Hrs.)<br><br>o Introduction to HTML<br>o HTML Elements and Structure<br>o Creating basic web pages with HTML<br>o Semantic HTML and accessibility<br><br>• Cascading Style Sheets (CSS) - (12Hrs.)<br>o Introduction to CSS<br>o CSS selectors and properties<br>o Styling web pages with CSS |

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| | • Introduction to Front-End development tools - (03Hrs.)<br>  ○ JavaScript<br>  ○ TypeScript<br>  ○ Next.JS<br><br>• JavaScript - (06Hrs.)<br>  ○ Introduction to JavaScript<br>  ○ Variables, Data Types, and Operators<br>  ○ Functions and Control Structures<br><br>• Introduction to Front-End Frameworks and Libraries - (03Hrs.)<br><br>• Website Hosting - (03Hrs.)<br>  ○ What is website hosting?<br>  ○ Types of Web Hosting<br>  ▪ Shared, VPS, Dedicated, and Cloud Hosting<br>  ○ Choosing the Right web hosting provider<br>  ○ Use of C-Panel<br><br>• Domain Names and DNS - (03Hrs.)<br>  ○ What is a Domain Name?<br>  ○ Domain Name Registration<br>  ○ Understanding DNS (Domain Name System)<br>  ○ Configuring DNS Settings<br><br>• Deploying a website - (03Hrs.)<br>  ○ Uploading website files to the hosting server<br>  ○ Basic server administration tasks |
| **Practical** | • Practical 01-Create a website with relevant navigations and interactions using HTML, CSS and JavaScript for a given scenario<br>• Practical 02 -Create a website with relevant navigations and interactions using web authoring tool / CMS for a given scenario<br>• Practical 03 - Host the created website |
| **Resources** | • Multimedia, Whiteboard<br>• Computer lab with Internet facility<br>• Smart board<br>• CMS- Ex: WordPress, Drupal, Joomla, etc.<br>• IDEs – Ex: VS Code, NetBeans, Eclipse, etc.<br>• XAMPP, WAMP |

<table>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>Head First HTML and CSS, 2nd Edition Elisabeth Robson, Eric Freeman ISBN: 9780596159900</li><li>Bootstrap: Responsive Web Development, 1st Edition Jake Spurlock ISBN: 9781449344597</li><li>Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 5th Edition Robin Nixon ISBN:978-1491978917</li><li>JavaScript: The Definitive Guide, 7th Edition David Flanagan ISBN: 9781491952023</li><li>Building Web Apps with WordPress, Second Edition, by Brian Messenlehner and Jason Coleman ISBN: 978-1-491-99008-7</li><li>Full Stack Web Development the Comprehensive Guide, Philip Ackermann, ISBN: 978-1-4932-2438-8</li></ul></td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3"><ul><li>Interactive Lecture with hands on practical sessions</li><li>Demonstration/practical</li><li>Individual assignment</li><li>Simulation</li><li>Programmed instructions</li></ul></td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Practical 01</td><td>30%</td></tr>
<tr><td>Practical 02</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam<br>(2 Hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T + 30 P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Creative design for print media** |
| **Module Code** | K72T001M08/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U05 |
| **Pre-Requisites** | Relevant NVQ Level 4 Qualification or GCE (A/L) or equivalent qualification |
| **Module Aim(s)** | **To enable the students to:**<br>• Prepare a creative design with text, graphics and images. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Design and develop graphics<br>2. Edit images<br>3. Finalize the text / graphic/images for printing |
| **Learning Content /Topics** | • Introduction to Text manipulation (02hrs.)<br>  ○ Font, Size, Formats, Special effects, Alignment, etc.<br>  ○ Text manipulation software<br>• Introduction to Graphic Designing (04hrs.)<br>  ○ Importance of Graphic designing<br>  ○ Sketching / Prototyping<br>  ○ Graphic Designing rules (Golden ratio, Gestalt theory)<br>  ○ Hardware and software requirement for installation<br>  ○ Graphic designing software<br>  ▪ Tools panel<br>  ▪ Layers<br>• Scanning (01hrs.)<br>  ○ Flat bed, Drum<br>  ○ OCR<br>• Introduction to Image editing (06hrs.)<br>  ○ Importance of Image editing<br>  ○ Camera shots and framing<br>  ○ Image editing standards<br>  ▪ Scaling<br>  ▪ Image quality<br>  ○ Hardware and software requirement for installation<br>  ○ Image editing software<br>  ▪ Tools panel<br>  ▪ Layers<br>• Page layout (02hrs.)<br>  ○ Page layout for different print media |

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| | ▪ Offset, Digital, etc.<br><br>• Types of graphics (02hrs.)<br>o Raster graphics<br>o Vector graphics<br><br>• Color theory (02hrs.)<br><br>• Use of Graphic Design software (02hrs.)<br>o Graphic Design Tools<br>o Packaging<br>o Import /Export Graphics /Images<br>o Slicing Graphics/Images<br><br>• Use of Image Editing software (06 hrs.)<br>o Image Editing Tools<br>o Packaging<br>o Import /Export Images<br>o Slicing Images<br><br>• Prepare Text/Images/Graphics for printing (01hrs.)<br>o Offset printing<br>o Digital printing<br>o Screen printing<br><br>• Ethical uses of Graphic and Images (02hrs.)<br>o Copyright Laws<br>o Plagiarism |
| Practical | Practical:<br>• Practical 01 - Prepare a graphic design for a given scenario (For example creating a poster with a logo)<br>• Practical 02 - Prepare a banner for a given scenario with images, graphics and texts<br>• Practical 03 – Prepare the inner page layouts of a magazine using InDesign<br>Assignment:<br>• Visit to a company and collect their requirement for printing and prepare a proposal /quotation for the above |
| Resources | • Relevant Software – Corel Draw, Illustrator, Photoshop, InDesign, etc.<br>• Multimedia, Whiteboard<br>• Computer lab with Internet facility<br>• Smart board |

<table>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>Designing for Print the Art and Science, Marina Poropat Joyce, ISBN: 978-0996214988</li><li>Layout Essentials Revised and Updated: 100 Design Principles for Using Grids, Tondreau, Beth, ISBN: 978-1-63159-631-5</li></ul></td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3"><ul><li>Interactive Lecture with hands on practical sessions</li><li>Demonstration/practical</li><li>Individual assignment</li><li>Simulation</li><li>Programmed instructions</li></ul></td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3"><ul><li>Lectures assisted with multimedia to deliver theory content</li><li>Demonstration</li><li>lectures</li><li>Practical (Individual)</li><li>Presentation</li><li>Project (Individual)</li></ul></td></tr>
<tr><td rowspan="7">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="4">Continuous Assessment</td><td>Practical 01</td><td>15%</td></tr>
<tr><td>Practical 02</td><td>15%</td></tr>
<tr><td>Practical 03</td><td>15%</td></tr>
<tr><td>Assignment:</td><td>15%</td></tr>
<tr><td rowspan="2">Summative Assessment</td><td>Prepare a poster/leaflet/label for a given scenario with images, graphics and texts</td><td>20%</td></tr>
<tr><td>Written Exam (01 hour)</td><td>20%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (30T + 60P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

# MODULE DESCRIPTORS
# Information and Communication Technology
# NVQ Level 05
## Semester I
## Employability modules

| | |
|---|---|
| **Module Title** | **Workplace Communications Management** |
| **Module Code** | EMTM02 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | EMTU02 |
| **Pre-Requisites** | None |
| **Module Aims** | **To enable the students to:**<br>• Apply communication skills to maintain effective workplace performance<br>• Adjust to diverse situations at workplace, through effective manipulation of communication skills |
| **Learning Outcomes** | **At The student will be able to:**<br>▪ Identify the essential components of an efficient communication system<br>▪ Identify and overcome barriers to effective communication<br>▪ Use all channels of communication equally well<br>▪ Use common computer applications to collect, analyze and maintain essential data and information required to perform and enhance day to day activities of the organization<br>▪ Contributes to the overall growth and productivity development of the organization |
| **Learning Content / Topics** | ▪ Basic communication models - (Reference to 2 typical models)<br>▪ Principles of effective, interactive communication<br>▪ Barriers to effective communication and distortions in the communication process<br>▪ Personal values and communication<br>▪ Policies of the organization relevant to information and communication function<br>▪ Protocol and Procedures of the organization<br>▪ Importance of ICT tools in promoting the efficiency and effectiveness of the organization<br>▪ Common computer applications<br>▪ Importance of networking in day-to-day activities of the organization |
| **Resources** | **Teaching Aids**<br>Networked Computers, Multimedia, Whiteboard |
| **Recommended Texts & / or References** | |
| **Teaching Learning Activities** | • Knowledge to be imparted by providing learner centered activities<br>• Facilitator may use different teaching methodologies such as brainstorming, projects, mind mapping, small group activities, illustrated talk, fish ball technique, demonstrations, when delivering knowledge |

<table>
<tr><td></td><td colspan="3">component of this module<br><br>The following may be discussed in groups of trainees and followed up with Exercises" & "Role plays"<br>• Oral communication<br>• Written communication<br><br>The following may be discussed in groups of trainees and followed up with" Exercises" & “Role plays"<br><br>• Constructing sound inductive arguments.<br>• Reading and comprehending written communications and information<br>• Using job-related terminology<br>• Using proper listening techniques</td></tr>
<tr><td rowspan="5">Assessment & Weighting</td><td>Type</td><td>Topic / Activity</td><td>Weighting</td></tr>
<tr><td rowspan="2">Formative assessment</td><td>Application of communication skills at work place</td><td>50%</td></tr>
<tr><td>Knowledge of various aspects of communication skills at work place - Oral questioning during class room presentations</td><td>10%</td></tr>
<tr><td rowspan="2">Summative assessment</td><td>Knowledge of various aspects of communication skills at work place- Multiple Choice Test Items/ Matching and Completion test items and structured essay type questions</td><td>30%</td></tr>
<tr><td>knowledge on application of communication skills- Viva voce</td><td>10%</td></tr>
<tr><td>Duration</td><td colspan="3">50 hours (30T + 15P + 05S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

# MODULE DESCRIPTORS
# Information and Communication Technology
# NVQ Level 05
## Semester 2
## Technical modules

| | |
|---|---|
| **Module Title** | **Software Testing & Quality Assurance** |
| **Module Code** | K72T001M09/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U04, K72T001U07 |
| **Pre-Requisites** | K72T001M01, K72T001M02, K72T001M05, K72T001M06, K72T001M07 |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Design and execute test cases<br>2. Use various testing tools for manual test<br>3. Conduct specialized tests such as performance, security |
| **Learning Content / Topics** | • Introduction to Software Testing – (12hrs.)<br>  ○ Definition and importance of software testing<br>  ○ Quality assurance vs. Quality control<br>  ○ Software testing lifecycle (STLC)<br>• Types of Testing (06hrs.)<br>  ○ Manual testing, Automated testing<br>  ○ Unit testing, integration testing, system testing, and acceptance testing<br>  ○ Functional vs. non-functional testing<br>• Testing Techniques and Strategies - (03hrs.)<br>  ○ Black-box testing techniques<br>  ○ White-box testing techniques<br>• Test Planning and Documentation - (15hrs.)<br>  ○ Test planning and strategy<br>  ○ Writing effective test cases<br>  ○ Test data preparation<br>  ○ Test documentation (test plan, test case, test script, test report)<br>• Introduction to Automation Testing - (03hrs.)<br>  ○ Introduction to test automation<br>  ○ Benefits<br>• Security Testing - (03hrs.)<br>  ○ Importance of security testing<br>  ○ Common security vulnerabilities<br>  ○ Introduction to tools and techniques for security testing |

<table>
<tr><td></td><td colspan="3"></td></tr>
<tr><td>Practical</td><td colspan="3">Assignment - Prepare the test documentation and present</td></tr>
<tr><td>Resources</td><td colspan="3">• Multimedia, Whiteboard<br>• Smart board<br>• Computers with relevant software</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Foundations of Software Testing ISTQB Certification, 4th edition Dorothy Graham, Rex Black, Erik van Veenendaal ISBN: 9780357884157<br>• Mastering Software Quality Assurance Murali Chemuturi ISBN: 9781604270327<br>• Software Quality Assurance Made Easy 2nd Edition Solis Tech [No ISBN]</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">• Interactive Lecture with hands on practical sessions<br>• Demonstration/practical<br>• Individual/Group assignment</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Assignment</td><td>30%</td></tr>
<tr><td>Practical</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Written Test (2 hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">75hrs. (42T + 18 P + 15 S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Web Application Development** |
| **Module Code** | K72T001M10/ Level 05 |
| **Module Type** | **Compulsory** |
| **Relevant Unit/s** | K72T001U07 |
| **Pre-Requisites** | K72T001M02, K72T001M07 |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Customize the theme/templates using a CMS<br>2. Add interactions to web pages using JavaScript<br>3. Connect a database to the web page/site<br>4. Integrate developed website with a payment gateway |
| **Learning Content / Topics** | • Cascading Style Sheets (CSS) - (06Hrs.)<br>  ○ CSS Positioning and Layouts<br>  ○ CSS Grid and Flexbox<br>• Responsive web design techniques - (03Hrs.)<br>• JavaScript - (03Hrs.)<br>  ○ Data validation<br>• DOM Manipulation (interacting with web page elements) - (09Hrs.)<br>  ○ Understanding the DOM<br>  ○ Selecting and Manipulating DOM Elements<br>  ○ Event Handling<br>  ○ Building user interfaces (UIs) with JavaScript<br>• Application of Front-End Frameworks and Libraries -(06Hrs.)<br>  ○ jQuery<br>  ○ Ajax<br>  ○ XML<br>• Back-End Programming - (12Hrs.)<br>  ○ Introduction to server-side programming concepts<br>  ▪ Client-Server Architecture<br>  ▪ Introduction to Server-Side Programming (back-end language/s)<br>  ▪ Setting Up a Local Development Environment<br>  ○ Introduction to Back-End development tools<br>  ▪ Node.JS<br>  ▪ Nest.JS<br>  ○ Introduction to back-end frameworks |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Databases - (12Hrs.)
<ul>
<li>Introduction to Databases</li>
<li>SQL vs NoSQL</li>
<li>Working with databases (CRUD operations)</li>
</ul></li>
<li>Website Security - (06Hrs.)
<ul>
<li>Common Web Security Threats</li>
<li>Securing the website from common threats</li>
<li>SSL certificates</li>
</ul></li>
<li>Introduction to payment gateway – (01hrs)</li>
<li>Customize the web hosting environment - (02Hrs.)</li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">
<ul>
<li>Practical 01 -Develop a responsive website with relevant navigations and interactions with a database connection for a given scenario</li>
<li>Practical 02 - Obtain a domain and publish the developed website</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Multimedia, Whiteboard</li>
<li>Smart board</li>
<li>Computer lab with internet facility</li>
<li>CMS- Ex: WordPress, Drupal, Joomla, etc.</li>
<li>IDEs – Ex: VS Code, NetBeans, Eclipse, etc.</li>
<li>XAMPP, WAMP</li>
<li>Low-Code No-Code tools and platforms</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>How To Build a Website, Dogit Al-Ju Domi, [No ISBN]</li>
<li>Learning Web Design - Html5 A Practical Handbook, Shanmugapriyaa S, ISBN: 978-93-5607-098-1</li>
<li>How to be a Web Developer, Radu Nicoara, ISBN: 978-1-4842-9663-9</li>
<li>Master Web Design with HTML CSS JavaScript and jQuery Create Stunning Interactive Websites, Frahann Hussain, Kamaron Hussain [No ISBN]</li>
<li>Java for Web Development, Sarika Agarwal, Vivek Gupta, ISBN: 978-93-55511-430</li>
</ul>
</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">
<ul>
<li>Interactive Lecture with hands on practical sessions</li>
<li>Demonstration/practical</li>
<li>Individual assignment</li>
<li>Simulation</li>
<li>Programmed instructions</li>
</ul>
</td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Practical 01</td><td>30%</td></tr>
<tr><td>Make a presentation and demonstration of practical 01 followed by viva (Individual)</td><td>10%</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | Practical 02 | 20% |
| | Summative Assessment | Written Exam (2 hour) | 40% |
| **Duration** | **100hrs. (60T + 30 P + 10 S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

| | |
|---|---|
| **Module Title** | **Database Management Systems** |
| **Module Code** | K72T001M11/ Level 05 |
| **Module Type** | Core |
| **Relevant Unit/s** | K72T001U03 |
| **Pre-Requisites** | None |
| **Duration (Hrs.)** | 100 hrs. |
| **Learning Outcomes** | After completion of this module the trainee will be able to:<br>1. Explain different database types<br>2. Develop Entity Relationships (ER) diagrams<br>3. Perform Normalization up to 3 NF<br>4. Mapping ER to Relational schema<br>5. Create required queries for a given scenario |
| **Learning Content** | ***Theory:***<br>● Introduction to Database Management Systems (DBMS) – (06hrs.)<br>o Compare manual file handling with computerized file handling<br>o The evolution of databases<br>o Database architectures (1-tier, 2-tier, 3-tier)<br>● Types of databases – (03hrs.)<br>o Hierarchical, Network, Relational, Object-oriented databases<br>o NoSQL, Cloud database<br>● Entity Relationship Diagrams (ERD) – (12hrs.)<br>o Components of ERD<br>▪ Entities – Strong, Weak<br>▪ Attributes – Key, Composite, Derived, Multi value, etc.<br>▪ Cardinality of the Relationship – 1: 1, 1:M, M: N<br>▪ Optionality of the relationship<br>▪ Degree of the relationship<br>o Mapping from ERD to Relational schema<br>● Fundamentals of Extended/Enhanced Entity Relationship Diagrams (EERD) – (06hrs.)<br>● Database Normalization – (12hrs.)<br>o Types of Anomalies<br>▪ Insert anomalies<br>▪ Delete anomalies<br>▪ Update anomalies |

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| | ○ Levels of Normalization<br>▪ 1st Normal Form (1st NF)<br>▪ 2nd Normal Form (2nd NF)<br>▪ 3rd Normal Form (3rd NF)<br>○ Database keys<br>▪ Super key, Candidate key, Primary key, Composite primary key, Foreign Key, Alternate key<br>● Structured Query Language (SQL) – (15hrs.)<br>○ Types of SQL statements<br>▪ Data Definition Language (DDL)<br>- Create, Alter, Drop, Truncate<br>▪ Data Manipulation Language (DML)<br>- Insert, Update, Delete, Select<br>▪ Data Control Language (DCL)<br>- Grant, Revoke<br>○ SQL Datatype<br>○ Constraints<br>▪ Unique, Primary key, foreign key, Not null, Check, Default<br>○ Create Queries<br>▪ JOIN and Sub Queries<br>○ Scaler functions and aggregated functions<br>○ Stored procedures, Functions and Triggers<br>● Backup and restore database /data – (03hrs.)<br>● Install and Configure DBMS – (03hrs.)<br>○ Access privileges |
| | ***Practical:***<br>• Practical 01 - Draw ER diagram for a given scenario<br>• Practical 02- Create relational schema for the drawn ER diagram<br>• Practical 03 - Make simple queries/ sub queries/ JOIN queries for the given scenarios |

<table>
<tr><td>Resources</td><td colspan="3">• Multimedia, Whiteboard<br>• Smart board<br>• Computer lab with internet facility<br>• XAMPP, WAMP, MySQL</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Introduction to Database Systems, 8th Edition C. J. Date, ISBN: 9780321197849<br>• Fundamentals of Database Management Systems, Third Edition, Mark L. Gillenson, ISBN: 978-1-119-90746-6<br>• Database Mastery with Python:SQL, NoSQL, and Beyond, László Bocsó, [No ISBN]<br>• Modern Database Management, JEFFREY A. HOFFER , V. RAMESH<br>• HEIKKI TOPI, ISBN 9780134773650<br>• Learning Relational Database, Shanmugapriyaa S ,ISBN: 978-93-5607-713-3<br>• Head First SQL: Your Brain on SQL - A Learner's Guide Lynn Beighley, ISBN: 9780596526849</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">• Interactive Lecture with hands on practical sessions<br>• Demonstration/practical<br>• Individual assignment<br>• Simulation<br>• Programmed instructions</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td rowspan="3">Continuous Assessment</td><td>Practical 01</td><td>10%</td></tr>
<tr><td>Practical 02</td><td>20%</td></tr>
<tr><td>Practical 03</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam</td><td>50%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T + 30 P + 10 S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Multimedia production** |
| **Module Code** | K72T001M12/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U06, K72T001U07 |
| **Pre-Requisites** | K72T001M08 |
| **Module Aim(s)** | To enable the students to:<br>• Understand multimedia concepts and its various elements (text, graphics, audio, video, animation). |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain multimedia production and its application in industries.<br>2. Explain principles of graphic design, layout, and typography.<br>3. Create graphics, audio, video, animation using relevant software tools<br>4. Understand and comply on media ethics and legal frameworks |
| **Learning Content / Topics** | • Introduction to multimedia (02 hrs.)<br>o Overview of multimedia and its types<br>o Multimedia in different industries (Education, Advertising, Gaming, etc.)<br>o Principles of design: Contrast, Alignment, Repetition, Proximity (C.R.A.P)<br>o Story boarding and concept development for multimedia projects<br>o Understanding user experience (UX) and user interface (UI) in multimedia<br><br>• Media ethics and legal frameworks (03 hrs.)<br>o Intellectual property and copyrights law<br>o Ethics in media<br>o Data protection and privacy<br>o Social media trends and regulations (Online safety acts)<br><br>• Graphics and Image Editing (08 hrs.)<br>o Introduction to Raster and Vector Graphics<br>o Raster and Vector graphic tools<br>o Editing Raster images<br>o Creating and manipulating vector images<br>o Color theory and image optimization for multimedia<br>o Designing for different media (web, print, video)<br>o Different graphic file formats |

| | |
|---|---|
| | • Audio Production and Editing (06 hrs.)<br>  ○ Sound theory: Frequency, pitch, volume<br>  ○ Audio recording techniques (using microphones, field recording)<br>  ○ Editing sound<br>  ○ Sound effects, Foley, and background music<br>  ○ Synchronizing audio with visual content (lip-syncing, video dubbing)<br>• Video Production and Editing (06 hrs.)<br>  ○ Video production: Camera operation, lighting, framing, angles and shots<br>  ○ Video editing principles (cutting, transitions, effects)<br>  ○ Motion graphics<br>  ○ Exporting videos for different platforms and file formats<br>• 2D Animation and Motion Graphics (03 hrs.)<br>  ○ Introduction to animation principles (squash & stretch, timing, anticipation)<br>  ○ Creating motion graphics for multimedia and web content<br>  ○ Integrating animation with video projects<br>• Introduction to visual projection, mapping and Playback (02 hrs.)<br>  ○ Live streaming<br>  ○ Live vision<br>  ○ LED wall mapping / projection<br>  ○ LED wall animation |
| **Practical** | • Practical 01 - Design a web poster<br>• Practical 02 - Create a soundscape for a short multimedia project<br>• Practical 03 - Create a minimum of 1-minute video montage with music and text<br>• Practical 04 - Live streaming on social media<br>• Practical 05 - Produce a short video with relevant titles<br>• Practical 06 - Create a 10-second animated logo or motion graphic<br>• Practical 07 - Animate a character or object using keyframing |

<table>
<tr><td>Resources</td><td colspan="3">• Internet connection<br>• Mobile phone or Device<br>• Video camera / Digital camera<br>• Lighting equipment<br>• Computer and related peripherals<br>• Required computer software<br>  • Photoshop<br>  • Illustrator<br>  • Corel Draw<br>  • Premier<br>  • After Effects<br>  • Audition<br>  • Vmix<br>  • Resolume Areena</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Zero to Mastery in Multimedia, Dr. R.K. Jain, [No ISBN: ]<br>• Multimedia Making it Work, Nineth Edition, Tay Vaughan, ISBN: 978-0-07-183288-5</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">• Lectures<br>• Demonstration<br>• Practical<br>• Assignment<br>• Case study<br>• Field visit to video wall production house/site/studio</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Complete a multimedia project (e.g., a promotional video, interactive website, short animation)</td><td>30%</td></tr>
<tr><td>Submit a portfolio with multimedia projects (video, audio, animation).</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Three (3) hour Practical test for a given scenario<br>Note: -<br>• Marking should be done just after the practical test<br>• Should assign maximum of fifteen candidates per invigilator<br>• Tested softcopies of the projects</td><td>40%</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | *and marking of each candidate should be stored in a storage media* | |
| **Duration** | **100hrs. (30T +60P + 10S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

| | |
|---|---|
| **Module Title** | **Computer Networks** |
| **Module Code** | K72T001M13/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U08 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Explain the requirement of computer networks<br>• Configure the network environment<br>• Segmentation the network |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Describe the important of computer network<br>2. Describe computer network devices.<br>3. Describe the different network types<br>4. Describe the client computer configuration and server configuration<br>5. Describe the important of the subnetting |
| **Learning Content / Topics** | ● Computer Network Fundamentals -(12Hrs.)<br>○ What is a computer Network<br>○ Advantages and Disadvantage of Networking<br>○ Network topologies<br>○ Types of Networks<br>▪ LAN, MAN, WAN, etc.<br>▪ Wired, Wireless<br>▪ Client-Server, Peer-to-Peer<br>○ Data communication fundamentals<br>▪ Data Transmission Modes – Simplex, Half-duplex, Full-duplex<br>▪ Transmission Media -Wired, Wireless<br>● Network devices -(06Hrs.)<br>○ Network switches<br>○ Routers<br>○ Repeaters<br>○ Media Converters<br>○ Access point<br>● Server computers -(03Hrs.)<br>○ Requirement of a server computer<br>○ Type of servers<br>▪ File, Print, Web, Proxy, DHCP, DNS, Database, etc.<br>● Network cables -(03Hrs.)<br>○ Cable types – Twisted, Co-axial, Fiber<br>• Network cabling tools -(01Hrs.) |

| | | | |
|---|---|---|---|
| | • Network testing tools -(01Hrs.)<br>• Computer network reference models -(02Hrs.)<br>  ○ OSI, TCP/IP<br>• Network Protocols -(02Hrs.)<br>• Addressing Host / Devices in a computer network -(03Hrs.)<br>  ○ Physical address (MAC Address)<br>  ○ Logical address (IP Address)<br>  ○ Dynamic, Static<br>  ○ IPv4, IPv6<br>  ○ Public, Private<br>• Subnetting -(09Hrs.)<br>  ○ Classful, Classless<br>• Installing and configuring network devices -(06Hrs.)<br>● Wireless Networking -(03Hrs.)<br>  ○ Introduction to wireless communication<br>  ○ wireless technologies (wi-fi, Bluetooth, wimax,2G,3G, etc.)<br>  ○ wireless security encryptions<br>  ○ Configuring wireless access points (WI-FI)<br>● Server Configuration -(09Hrs.)<br>  ○ Server Operating System Types<br>  ○ Domain/Workgroup Configuration<br>  ○ Proxy, Firewall configuration | | |
| **Practical** | Practical:<br>● Practical 01 - Create a subnet for a given scenario<br>● Practical 02 - Install and setup a file server/web server / proxy | | |
| **Prescribed Texts & / or References** | • Computer Networking Bible, Kevin Sims, 2025 Edition, [No ISBN]<br>• Computer Networks, Sixth Edition, Andrew S. Tanenbaum, ISBN: 978-93-560-6325-9<br>• Networking All-In-One, 9th Edition, Doug Lowe, ISBN: 978-1-394-27839-8<br>• Networking for Beginners, Jack Mathew, [No ISBN] | | |
| **Resources** | • Network switches<br>• Routers<br>• Access point<br>• Network toolbox<br>• Multimedia, Whiteboard<br>• Smart board<br>• Computer lab with internet facility | | |
| **Assessment &** | **Type** | **Topic / Activity/** | **Module Weighting** |

| | | | |
|---|---|---|---|
| **Weighting** | Continuous Assessment | Practical 01 | 30% |
| | | Practical 02 | 30% |
| | Summative Assessment | Written Exam (3 hour) | 40% |
| **Duration** | **100hrs. (60T + 30 P + 10 S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

| | |
|---|---|
| **Module Title** | **Cyber Security and Cloud computing** |
| **Module Code** | K72T001M14/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U08 |
| **Pre-Requisites** | K72T001M03, K72T001M13 |
| **Module Aim/s** | To enable the students to:<br>● Explain the important of the protection of data and privacy from cyber attack<br>● Explain the different types of cyber attacks<br>● Explain how to protect from cyber attacks<br>● Explain the cloud services provided by different service providers |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Describe the CIA triad<br>2. Describe the common cyber security threats<br>3. Describe the countermeasures for cyber attacks<br>4. Describe the important of cloud computing services |
| **Learning Content / Topics** | • Introduction to Cybersecurity -(04Hrs.)<br>o Describe the principles of confidentiality, integrity, and availability (CIA Triad)<br>o Compare how cybersecurity threats affect individuals, businesses, and organizations<br>o Types of cybersecurity countermeasures<br>• Cybersecurity Threats, Vulnerabilities and Attacks -(04Hrs.)<br>o Malware and Malicious Code<br>o social engineering<br>o Attacks<br>• Protecting Your Data and Privacy -(04Hrs.)<br>o protect confidentiality.<br>▪ Cryptography<br>▪ Access Controls<br>▪ Obscuring Data<br>o Data Integrity<br>▪ Digital Signatures<br>▪ Data Encryption<br>▪ Certificates<br>o High Availability<br>▪ Disaster Recovery<br>▪ Data Redundancy<br>• Introduction to cloud computing (06Hrs.)<br>o Cloud services (IaaS, PaaS, SaaS) |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Cloud service providers and their services</li>
<li>Cloud deployment models</li>
</ul>
<ul>
<li>Accessing cloud services -(03Hrs.)</li>
<li>Configuration of cloud services -(03Hrs.)</li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">
<ul>
<li>Assignment 01: - Create an analysis report for three cyber security attacks happened recently</li>
<li>Assignment 02: - Create a report to compare and contrast the services provided by different cloud service providers</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Multimedia, Whiteboard</li>
<li>Smart board</li>
<li>Computer lab with internet facility</li>
<li>Cyber Security Software Tools (Kali Linux, Wireshark, Nmap)</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>Cloud Computing Concepts, Technology, Security & Architecture, SECOND EDITION, Thomas Erl, Eric Barceló Monroy, ISBN: 978-0-13-805225-6</li>
<li>Cloud Computing From Beginning to End, Ray Rafaels, ISBN: 978-1986726283</li>
<li>CYBER SECURITY A practitioner’s guide, David Sutton, ISBN: 978-1-78017-342-9</li>
<li>The Cybersecurity and Computer Networking Bible, Trevor Shelwick, [No ISBN]</li>
<li>Cybersecurity All-in-One For Dummies, Joseph Steinberg; Kevin Beaver; Ted Coombs; and Ira Winkler, ISBN: 978-1-394-15286-5</li>
<li>The Complete Cyber Security Course Volume I HACKERS Exposed, Nathan House, [No ISBN]</li>
</ul>
</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/</td><td>Module Weighting</td></tr>
<tr><td rowspan="3">Continuous Assessment</td><td>Assignment 01</td><td>20%</td></tr>
<tr><td>Assignment 02</td><td>20%</td></tr>
<tr><td>Presentation for a selected cyber security concept including the cloud services</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (2 hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">50hrs. (24T + 18 P + 08 S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Developing Mobile App** |
| **Module Code** | K72T001M15/ Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U09 |
| **Pre-Requisites** | K72T001M01, K72T001M02, K72T001M05, K72T001M06 |
| **Module Aim(s)** | To enable the students to:<br>• Understand the mobile application development technologies and develop a comprehensive mobile application using a chosen technology |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>• Explain the strengths and weaknesses of each technology<br>• Develop mobile applications using native, web native, and hybrid technologies<br>• Test the developed mobile application |
| **Learning Content / Topics** | • Introduction to Mobile Application Development (03 hrs)<br>o Overview of Mobile Application Development<br>o Types of mobile applications (native, web native, hybrid)<br>o Mobile app development frameworks and tools<br>o Mobile Operating Systems (iOS, Android, etc.)<br>o Setting up Development Environments<br><br>• Native Mobile App Development (20 hrs)<br>o iOS Development<br>o Swift Programming Language<br>▪ Variables, Functions, Control Flow, OOP Concepts<br>o Introduction to Xcode IDE<br>o iOS UI Components & Design Patterns<br>▪ UIKit and SwiftUI<br>▪ Auto Layout, Storyboards, and Interface Builder<br>o Navigation and Data Persistence<br>▪ Core Data, User Defaults, File System<br>o Testing and Debugging iOS Apps<br>o Deployment of iOS Apps<br><br>o Android Development<br>o Kotlin Programming Language<br>▪ Variables, Functions, Control Flow, OOP Concepts |

| | |
|---|---|
| | ○ Introduction to Android Studio IDE<br>○ Android UI Components & Design Patterns<br>▪ Layouts, Fragments, Recycler View, Jetpack Compose<br>○ Navigation and Data Persistence<br>▪ SQLite, Room Database, Shared Preferences<br>○ Testing and Debugging Android Apps<br>○ Deployment of Android Apps<br><br>• Web-Native Development (12 hrs)<br>○ Introduction to Web-Native Applications<br>○ Progressive Web Apps (PWAs) using HTML, CSS, and JavaScript<br>○ Testing and Deploying PWAs<br>○ Creating a responsive web app<br><br>• Hybrid Mobile App Development (05 hrs.)<br>○ Introduction to Hybrid Frameworks<br>○ React Native<br>▪ React Native and Component Structure<br>▪ Accessing Native Features with React Native<br>▪ Deploying React Native Apps<br>○ Flutter<br>▪ Dart Programming Language<br>▪ Widgets, State Management, and Navigation<br>▪ Accessing Native Features through Platform Channels<br>▪ Testing and Deployment<br><br>• Backend Integration and API Development (08 hrs)<br>○ Backend for Mobile Apps<br>○ Building a Backend<br><br>• Performance Optimization (02 hrs)<br>○ Memory Management, Reducing App Size, Improving Load Time |
| **Practical** | • Practical 01: Develop a To-Do List App where users can:<br>○ Add new tasks.<br>○ Mark tasks as completed or not completed.<br>○ Delete tasks.<br>○ View a list of pending and completed tasks.<br><br>• Practical 02: Develop a Contact Book App where users can:<br>○ Add new contacts.<br>○ View a list of saved contacts.<br>○ Search for contacts by name. |

<table>
<tr><td></td><td colspan="3">o Edit and delete contacts.</td></tr>
<tr><td>Resources</td><td colspan="3">• Internet connection<br>• Mobile phone or Device<br>• Computer and related peripherals<br>• Required computer software</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Mastering Mobile Application Development, 2nd Edition, Jonathan Engelsma and Hans Dulimarta, [No ISBN]</td></tr>
<tr><td>Teaching/ LearningActivities</td><td colspan="3">• Lectures<br>• Demonstration<br>• Practical<br>• Assignment<br>• Case study</td></tr>
<tr><td rowspan="5">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="3">Continual Assessment</td><td>Practical 01</td><td>15%</td></tr>
<tr><td>Practical 02</td><td>15%</td></tr>
<tr><td>• Develop an Event Planner App where users can:<br>o Create new events with details like name, date, time, and description.<br>o View a list of upcoming events.<br>o Edit or delete existing events.<br>o Search for events by name or date.<br>o Set reminders or notifications for events.</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Three (3) hour Practical test for a given scenario<br>Note: -<br>• Marking should be done just after the practical test<br>• Should assign maximum of fifteen candidates per invigilator<br>• Tested apps and marking of each candidate should be stored in a storage media</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">75hrs. (50T + 20P + 05S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

# MODULE DESCRIPTORS
# Information and Communication Technology
# NVQ Level 05
## Semester 2
## Employability modules

| | |
|---|---|
| **Module Title** | **Workplace Information Management** |
| **Module Code** | EMTM01/Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | EMTU01 |
| **Pre-Requisites** | None |
| **Module Aims** | **To enable the students to:**<br>• Make use of information and information systems to carry out organizational functions<br>• Make use of information and information systems to enhance workplace performance |
| **Learning Outcomes** | **At the end of this module the student shall be able to:**<br>• Explain the importance of ICT tools in promoting the productivity of the organization<br>• Describes the purpose of establishing ICT tools and strategies in enhancing the performance at workplace<br>• Apply Information skills to enhance the productivity of the organization |
| **Learning Content / Topics** | • Identification of documentation requirements<br>• Selecting and/or collecting required documentation<br>• Documentation procedures and methods<br>• Completing /perfecting documentation<br>• Reading, interpreting and using equipment/system manuals and specifications<br>• Interpretation of all applicable laws, policies and procedures relevant to enterprise<br>• Computer and information system usage<br>• Forecasting Techniques<br>• Forecasting Software<br>• Mathematical Modeling<br>• Data Collection Techniques for Market Research<br>• The range of analytical techniques appropriate for analysis of information<br>• The influence of human factors on information analysis, e.g. Prejudices and biases |

<table>
<tr><td></td><td colspan="3">• Conducting and recording of performance evaluations</td></tr>
<tr><td>Resources</td><td colspan="3">Teaching Aids<br>Networked Computers, Multimedia, Whiteboard</td></tr>
<tr><td>Recommended Texts & / or References</td><td colspan="3">Information Management: Best Practices – Volume 1<br>(Bob Boiko, Erik M. Hartman)</td></tr>
<tr><td>Teaching Learning Activities</td><td colspan="3">• Discuss/ explain and provide essential theoretical inputs.<br>• Emphasize the importance of a valid information system in promoting customer relations - make reference to following<br>- Identification of customer needs.<br>- Measurement of customer needs and satisfaction.<br>- Obtaining feedback from customers.<br>- Recognition and understanding of customer problems and resolution or timely referral of problem in a manner satisfactory to the customer.<br>▪ Application of enterprise policies in satisfying customer needs<br>▪ Identifying enterprise Protocols associated with "Customer Services"<br>• Discuss "Satisfying customer complaints" using information sources available</td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity</td><td>Weighting</td></tr>
<tr><td rowspan="2">Formative assessment</td><td>Use of ICT Tools, Software applications, application of information management skills at work place</td><td>50%</td></tr>
<tr><td>Knowledge on various aspects of information management skills at work place - Oral questioning during class room presentations</td><td>10%</td></tr>
<tr><td>Summative assessment</td><td>Knowledge on various aspects of information management skills at work place - Multiple</td><td>30%</td></tr>
</table>

| | | | |
|---|---|---|---|
| | | Choice Test Items/ Matching and Completion test items and structured essay type questions | |
| | | knowledge on application of information management skills- Viva voce | 10% |
| **Duration** | 75 hours **(45T + 20P + 10S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

| | |
|---|---|
| **Module Title** | **Planning scheduling work at workplace** |
| **Module Code** | EMTM03/Level 05 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | EMTU03 |
| **Pre-Requisites** | None |
| **Module Aims** | **To enable the students to:**<br>• Plan and schedule work to be performed at workplace<br>• Assign work to workers based on assessment of competencies / work capacities of individual workers/working team<br>• Predict likely problems / probable changes that would come up in implementation of planned schedule |
| **Learning Outcomes** | **At the end of this module the student shall be able to:**<br>• Prepare a work schedule for a given work situation<br>• Determine work priorities as per predetermined criteria such as goals, targets and organizational requirements<br>• Develop a plan (process) to complete work to be done in a given situation<br>• Assess competencies of individual workers before assigning work |
| **Learning Content / Topics** | • Goals and Objectives of the organization<br>• Planning priorities<br>• Plans related to work and related activities at workplace<br>• Systems, procedures and processes relevant to the organization<br>• Quality and continuous improvement processes applied within the organization<br>• Company specific performance standards<br>• Industry/Workplace Codes of Practice /Codes of ethics<br>• Frontline management roles applicable to team management<br>• Manufacturer' s specifications and product specifications<br>• Standard specifications of commonly used materials<br>• Simple planning techniques/methods - (two to three common techniques/methods) |

<table>
<tr><td></td><td colspan="3">• Forecasting Techniques /methods<br>• Time management techniques<br>• Competency assessment methods</td></tr>
<tr><td>Resources</td><td colspan="3">Teaching Aids<br>Networked Computers, Multimedia, Whiteboard</td></tr>
<tr><td>Recommended Texts & / or References</td><td colspan="3"></td></tr>
<tr><td>Teaching Learning Activities</td><td colspan="3">• Small group activity -1 - Identify work requirements of a drawing office or of a given project<br>• Small group activity -2 - Set work priorities as per organizational requirements goals and targets<br>• Small group activity -3 -Develop a plan (process) to complete work to be done at drawing office or in a given project<br>• Individual activity - Prepare list/s of tools, equipment, material required to complete the work as indicated in the plan developed in activity -3<br>• Small group activity -4 Brainstorm in groups to identify,<br>• Deficiencies in the plan (Developed in activity -3)<br>• Modifications to be made to overcome deficiencies<br>• Explain and follow up with exercises<br>• Forecasting methods / Techniques, Time management techniques, Competency assessment method</td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity</td><td>Weighting</td></tr>
<tr><td rowspan="2">Formative assessment</td><td>Use of different approaches and techniques related to planning of work, application of skills in Planning work at work place</td><td>50%</td></tr>
<tr><td>Knowledge of various aspects of planning of work to be performed at workplace - Oral questioning during class room presentations</td><td>10%</td></tr>
</table>

| | | | |
|---|---|---|---|
| | Summative assessment | Knowledge of various aspects of planning of work to be performed at workplace - Multiple Choice Test Items/ Matching and Completion test items and structured essay type questions | 30% |
| | | knowledge on application of planning of work in a given work situation - Viva voce | 10% |
| **Duration** | 75hours **(45T + 20P + 10S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

# MODULE DESCRIPTORS
# Information and Communication Technology
# NVQ Level 06
## Semester I
## Technical modules

| | |
|---|---|
| **Module Title** | **IoT Development** |
| **Module Code** | K72T001M16/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U15 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Understand the fundamentals of electronic<br>• Understand fundamental of IoT development<br>• Familiarize with software systems and IoT integration |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>3. Demonstrate IoT development process<br>4. Explore functions of sensors and actuators<br>5. Develop IoT applications using relevant IoT programming language/s<br>6. Test and finalize developed IoT application |
| **Learning Content / Topics** | • Principles of Electricity – (4hrs.)<br>o Current, voltage, resistance<br>o Ohm's law<br>o Units & measurement<br>o Electrical equations<br>• Introduction to electronics – (4hrs.)<br>• Electronic components – (6hrs.)<br>o Resisters<br>o Transistors<br>o Capacitors<br>o Diodes<br>o Inductors<br>o Transformers<br>o ICs<br>o Relays<br>• Arduino Programming – (36 hrs.)<br>o Arduino development boards<br>o Components of Arduino development<br>o Arduino IDE<br>o Arduino Programming Languages<br>o Arduino programming projects<br>• Other IoT development boards– (10hrs.)<br>o Features and functions |
| **Practical** | • Perform following activities using Arduino board<br>o Blinking LED/s |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Traffic light system</li>
<li>Knight Rider LED Program Using 5 LEDs</li>
<li>Running a DC motor</li>
<li>Controlling a Servo Motor</li>
<li>Finding distance for an object using Ultrasonic Sensor</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Computer</li>
<li>Internet Facility</li>
<li>Relevant Software</li>
<li>Relevant Arduino and Electronic Components</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>Introduction to Internet of Things (IoT), Ahmed Banafa, ISBN: 978-1-003-42624-0</li>
<li>Introduction to IoT, Sudip Misra, Anandarup Mukherjee, Arijit Roy, ISBN: 978-1-108-95974-2</li>
</ul>
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
<ul>
<li>Lectures</li>
<li>Presentations</li>
<li>Demonstration</li>
<li>Discussions using the materials/ components and product samples</li>
</ul>
</td></tr>
<tr><td rowspan="5">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="3">Continuous Assessment</td><td>Playing music /buzzer according to the distance of an object using an ultrasonic sensor (Group – maximum five members)</td><td>25%</td></tr>
<tr><td>Detect a movement of an object using a motion sensor (Group – maximum five members)</td><td>25%</td></tr>
<tr><td>Controlling four traffic lights on roundabout (Individual)</td><td>25%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (01 hour)</td><td>25%</td></tr>
<tr><td>Duration</td><td colspan="3">150hrs. (60T + 60P + 30S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Project Management** |
| **Module Code** | K72T001M17/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U14 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Familiarize with project management techniques and tools |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>• Explain phases of the project management lifecycle<br>• Explain the activities performing in each phase<br>• Apply relevant project management techniques and tools<br>• Prepare project report |
| **Learning Content / Topics** | • Project Management concepts – (5 hrs.)<br>o Definition of Project Management<br>o Roles of a project manager<br>o Importance of Project Management<br>o Project management life cycles<br><br>• Project initiation – (5 hrs.)<br>o Project Charter<br>o Identifying Key Stakeholders<br>o Project feasibility<br><br>• Project Planning – (15 hrs.)<br>o Project Scope<br>o Project schedules (WBS, Gantt chart, etc.)<br>o Resource Allocation<br>o Budgeting and Cost Estimation<br>o Risk Management and Mitigation Strategies<br><br>• Project execution – (10 hrs.)<br>o Agile Project Management<br>- Introduction to Agile Methodologies (Scrum, Kanban, etc.)<br>- Agile Principles and Values (Manifesto for Agile Software Development)<br>- Sprints and Iterations in Agile<br>- Roles in Agile (Product Owner, Scrum Master, Development Team, etc.) |

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| | ○ Waterfall vs. Agile Methodologies<br>- Overview and Comparison of Waterfall and Agile<br>○ Team Management<br>- Communication and Collaboration in Cross-Functional Teams<br>- Conflict resolution<br>- Managing Stakeholders<br>• Project Monitoring and Control – (10 hrs.)<br>○ Tracking Progress Using Key Performance Indicators (KPIs)<br>○ Managing Project Scope, Time, and Budget Constraints<br>○ Risk management<br>○ Change Control Management<br>○ Feedback management<br>• Project Closure –(05 hrs.)<br>○ Project evaluation<br>○ Project Closure Report<br>• Project Management Tools – (10 hrs.)<br>○ Tools for Planning, Scheduling, and Task Management (Jira, Trello, Asana, etc.)<br>○ Time Tracking and Productivity Tools |
| **Practical** | • Assignment: Prepare a project report for the software project including,<br>○ Project charter<br>○ Project scope<br>○ Project feasibility<br>○ Project schedules<br>○ Resource allocation<br>○ Budgeting and Cost Estimation<br>○ Risk factors<br>○ Project Closure |
| **Resources** | • Computer<br>• Internet Facility<br>• Relevant Software |
| **Prescribed Texts & / or References** | • PMBOK Seventh Edition, Project Management Institute, ISBN: 978-1-62825-664-2<br>• Software Project Management Methods and Techniques, Lawrence J. Peters, ISBN: 978-1-003-48428-8<br>• The Handbook of Project Management, Sixth Edition, Martina Huemann, Rodney Turner, ISBN: 978-1-003-27417-9<br>• The Non-Project Manager's Guide to Project Management, Sam Buah, [No ISBN] |

<table>
<tr><td></td><td colspan="3"><ul><li>Green Project Management, Richard Maltzman, David Shirley, ISBN 978-1-4398-3001-7</li></ul></td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3"><ul><li>Lectures</li><li>Presentations</li><li>Demonstration</li><li>Practical</li><li>Discussions using the material</li></ul></td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Assignment</td><td>40%</td></tr>
<tr><td>Make presentation of the prepared project report</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (2 hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T + 30P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Relational and NoSQL Databases** |
| **Module Code** | K72T001M18/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U11 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Explain the advantages and uses of NoSQL Databases |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Select the appropriate NoSQL database<br>2. Configure the NoSQL database<br>3. Manage Insert, Update, Delete and Query dataset with NoSQL database<br>4. Optimize the NoSQL database<br>5. Backup and restore the NoSQL database |
| **Learning Content / Topics** | • Revision of Relational database concepts (01 hour)<br>• Relational Vs Non-relational databases – (2hrs)<br>• Introduction to NoSQL – (3hrs)<br>• NoSQL data models– (5hrs)<br>  ○ Key-value<br>  ○ Column family<br>  ○ Document<br>  ○ Graph<br>• Setup and Configuration – (14hrs)<br>  ○ Features of NoSQL databases<br>  i. Key-Value<br>  ii. Document<br>  iii. Column Family<br>  iv. Graph<br>  ○ Installation of NoSQL Database<br>  ○ Configuring NoSQL Database<br>• Database Connectivity– (5hrs)<br>  ○ Command-line interfaces (CLI) and Client tools<br>  ○ Drivers for programming languages<br>• Distribution Models – (4hrs)<br>  ○ Single Server<br>  ○ Sharding<br>  ○ Master-Slave Replication<br>  ○ Peer-to-Peer Replication<br>• Query NoSQL database– (20hrs)<br>• CRUD operations– (10hrs) |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Create</li>
<li>Read</li>
<li>Update</li>
<li>Delete</li>
</ul>
<ul>
<li>Backup and restore (4hrs)</li>
<li>Performance Monitoring and Tuning– (6hrs)
<ul>
<li>Indexing Strategies</li>
<li>Denormalization Techniques in NoSQL</li>
<li>Caching Mechanisms</li>
</ul></li>
<li>Security in NoSQL Databases– (10hrs)
<ul>
<li>Authentication and Authorization</li>
<li>Encryption of Data</li>
<li>Common Security Vulnerabilities</li>
</ul></li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">
<ul>
<li>Configure given NoSQL database system</li>
<li>Perform CRUD operations for a given scenario</li>
<li>Write and execute Queries for a given scenario</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Computer</li>
<li>Internet Facility</li>
<li>Relevant Software (Redis, MongoDB, Cassandra, Neo4j, etc.)</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>Seven NoSQL Databases in a Week: Get up and running with the fundamentals and functionalities of seven of the most popular NoSQL databases by Aaron Ploetz (Author), Devram Kandhare (Author), Sudarshan Kadambi (Author), Xun (Brian) Wu (Author)</li>
</ul>
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
<ul>
<li>Lectures</li>
<li>Presentations</li>
<li>Demonstration</li>
<li>Practical</li>
<li>Discussions using the material</li>
</ul>
</td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td>Continuous Assessment</td><td>Practical: create database, perform CRUD operation and execute queries for a given scenario</td><td>60%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (2 hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">150hrs. (84T + 36P + 30S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Information Security concepts and Ethics** |
| **Module Code** | K72T001M19/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U24, K72T001U25, K72T001U26, K72T001U27 |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>Provide comprehensive knowledge of information security standards, the formulation and implementation of security policies, procedures, legal, regulatory, and ethical frameworks to govern information security |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Explain key information security standards and frameworks<br>2. Develop and enforce organizational security policies and procedures<br>3. Interpret and comply with relevant security laws and regulations<br>4. Explain compliance requirements for organizations<br>5. Apply ethical principles in information security decision-making |
| **Learning Content / Topics** | • Information Security Standards (09 hrs.)<br>o Overview of Information Security Standards and frameworks (e.g., ISO 27001, NIST Cybersecurity Framework, COBIT etc.)<br>o The Role of Standards in Information Security Management<br>o Importance of Compliance and Certification<br>o Auditing and certification processes for security standards<br><br>• Security Policies, Procedures, and Guidelines (09 hrs.)<br>o Key Elements of Information Security Policies<br>o Policy Development Process (Assessment, Drafting, Approval, Implementation)<br>o Access control policies and procedures<br>o Data classification and handling procedures<br>o Incident response policies and procedures<br>o Security awareness and training programs<br>o Communication and Training on Security Policies<br><br>• Security Laws, Regulations, and Compliance (9 hrs.)<br>o Key cybersecurity laws and regulations (e.g. –Online Safety Act Sri Lanka, etc.)<br>o Data privacy and protection laws (e.g. -Data Protect Act Sri Lanka, etc.)<br>o Intellectual property rights and cybersecurity<br>o Legal and ethical considerations in cybersecurity |

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|---|---|
| | • Risk Management and Compliance (6 hrs.)<br>  ○ Risk assessment and management methodologies<br>  ○ Threat modeling and vulnerability analysis<br>  ○ Continuous monitoring and improvement of security controls<br><br>• Ethical Considerations in Information Security (6 hrs.)<br>  ○ Overview of Ethics in Information Security<br>  ○ Ethics vs. Legal Compliance<br>  ○ Professional Codes of Conduct (e.g. ISC², ISACA, etc.)<br>  ○ Responsibilities of Security Professionals<br>  ▪ Job description and work scope of authorized personnel to the system<br>• Emerging Trends and Technologies (9 hrs.)<br>  ○ Cloud security and compliance<br>  ○ Internet of Things (IoT) security and compliance<br>  ○ Artificial intelligence (AI) and machine learning in cybersecurity<br>  ○ Blockchain technology and its impact on security |
| **Practical** | • Practical 1: Develop an information security policy document for a small-scale organization<br>• Practical 2: Prepare a report on a given case study about information/cyber security scenario<br>• Practical 3: Prepare a report on by critically analyzing a given information security/cyber security policy document (report should include demerits of the current policy document and ways to improve/enhance them)<br>• Practical 4: Analyze a real-world case study involving security law violations |
| **Resources** | • Computer<br>• Internet Facility<br>• Relevant Software |
| **Prescribed Texts & / or References** | • Computer and Information Security Handbook - 3rd Edition, John R. Vacca<br>• The Complete Cyber Security Course, Volume 1 Hackers Exposed, Nathan House<br>• Principles of Information Security - 5th Edition, Whitman, Michael Mattord, Herbert<br>• Foundations of Information Security A Straightforward Introduction, Jason Andress<br>• Cybersecurity for Dummies, Joseph Steinberg<br>• Network Security For Dummies, 1st Edition, Chey Cobb<br>• CEH v10 Certified Ethical Hacker Study Guide, by Ric Messier<br>• CEH Certified Ethical Hacker All-in-One Exam Guide, Author Matthew Walker |

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|---|---|---|---|
| **Teaching/Learning Activities** | • Lectures, Presentations, Demonstration, Practical, Discussions using the material | | |
| **Assessment & Weighting** | **Type** | **Topic / Activity/Learning outcomes** | **Module Weighting** |
| | Continuous Assessment | Practical 01 | 10% |
| | | Practical 02 | 10% |
| | | Practical 03 | 10% |
| | | Practical 04 | 10% |
| | | Make two presentations based on two of the above practical | 20% |
| | Summative Assessment | Written Exam -2 hours | 40% |
| **Duration** | **100hrs. (48T + 30P + 22S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

| | |
|---|---|
| **Module Title** | **Manage Secured Mobile App Development** |
| **Module Code** | K72T001M20/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U13 |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>Equip learners with the knowledge, skills, and competencies to design, develop, and manage secured mobile applications across various platforms, ensuring robust security practices are integrated at every stage of the development lifecycle |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Explain the principles of mobile application security and the threats facing mobile devices and applications<br>2. Explain the mobile application development lifecycle and the role of security at each stage<br>3. Explain the principles of cryptography and how they are applied in mobile security<br>4. Conduct penetration testing of mobile applications |
| **Learning Content / Topics** | • Mobile App Security Fundamentals (12 hrs.)<br>o Mobile platforms and architectures (e.g. iOS, Android)<br>o Common mobile app vulnerabilities (e.g., data breaches, malware, phishing, etc.)<br>o Threat modeling and risk assessment methodologies<br>o Mobile app security standards and compliance (e.g., OWASP Mobile Security Testing Guide, GDPR, etc.)<br>o Legal and ethical considerations in mobile app security<br><br>• Secure Mobile App Development Lifecycle (12 hrs.)<br>o Secure design principles and patterns<br>o Secure coding practices for mobile platforms<br>o Code reviews and security testing techniques<br>o Static and dynamic application security testing (e.g. SAST, DAST, etc.)<br>o Penetration testing of mobile apps<br>o Secure deployment and release management<br><br>• Cryptography and Data Protection (12 hrs.)<br>o Cryptographic algorithms and protocols (e.g., encryption, hashing, digital signatures)<br>o Secure data storage and transmission techniques<br>o Authentication and authorization mechanisms (e.g., biometrics, multi-factor authentication, etc.) |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Data privacy and compliance regulations</li>
</ul>
<ul>
<li>Mobile Device Security (12 hrs.)
<ul>
<li>Mobile device management (MDM) and enterprise mobility management (EMM)</li>
<li>Secure mobile device configuration and hardening</li>
<li>Mobile threat defense (MTD) solutions</li>
<li>BYOD (Bring Your Own Device) security considerations</li>
<li></li>
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</li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">Practical: Develop one of the following Mobile App<br>1. Secure Mobile Banking App<br>2. Secure Healthcare/Health Tracking App<br>3. Secure E-commerce App<br>4. Secure Supply Chain App<br>5. Secure Logistics Management App</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Computer</li>
<li>Internet Facility</li>
<li>Relevant Software</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>Mobile Application Security: Protecting Mobile Devices and Their Applications, Himanshu Dwivedi; Chris Clark; David Thiel; Dawsonera, ISBN: 9780071633574</li>
</ul>
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
<ul>
<li>Lectures</li>
<li>Presentations</li>
<li>Demonstration</li>
<li>Practical</li>
<li>Discussions using the material</li>
</ul>
</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Practical</td><td>40%</td></tr>
<tr><td>Make a presentation on the Secure Mobile App Developed</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (2 hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (48T + 48P + 04S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Green IT Concept** |
| **Module Code** | K72T001M21 Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | All Units |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>Empowers learners to reduce the energy use, waste, and other environmental impacts of IT systems while reducing life cycle costs, thereby improving competitive advantage and learn how to procure sustainable hardware, design green data centers, recycle computer equipment, configure computers to minimize power, use virtualization to reduce the number of servers, and other green technologies, make green IT an integral part of organizational culture and planning. |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Explain the principles and significance of Green IT<br>2. Evaluate the environmental and economic impact of IT systems<br>3. Implement methodologies to reduce energy consumption and e-waste in IT operations<br>4. Apply Green IT principles |
| **Learning Content / Topics** | • Green IT (12 hrs.)<br>o Definition and importance of Green IT<br>o Environmental impact of IT<br>▪ E-waste, energy consumption, resource depletion, carbon footprint<br>▪ Paperless office concept<br>o Regulatory frameworks and compliance<br>▪ ISO 14001: Environmental Management System<br>▪ WEEE Directive (Waste Electrical and Electronic Equipment Directive)<br>▪ LEED (Leadership in Energy and Environmental Design) for IT<br>▪ The Green Grid: Industry standards and best practices<br>▪ Energy Star<br>▪ E-waste disposal regulations<br>o Benefits of adopting Green IT practices<br><br>• Energy Efficiency in IT Infrastructure (09 hrs.)<br>o Energy-efficient hardware, Software and components<br>▪ Selecting energy-efficient hardware components (processors, memory, storage)<br>▪ Implementing power management features in operating |

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|---|---|
| | systems and devices<br>▪ Green software design and development practices<br>o Power management techniques in data centers and servers<br>▪ Energy-intensive components in data centers (servers, storage, cooling)<br>o Virtualization and consolidation for energy savings<br>▪ Benefits of virtualization for energy efficiency and resource optimization<br>▪ Server consolidation and virtualization technologies<br>o Energy-efficient cooling solutions for IT facilities<br>▪ Energy-efficient cooling strategies (air cooling, water cooling)<br><br>• Sustainable Software Development Practices (06 hrs.)<br>o Green software design and development practices<br>o Green coding practices and tools<br><br>• E-Waste Management and Recycling (12 hrs.)<br>o Types and composition of e-waste<br>o E-waste problem and its environmental and social impacts<br>o E-waste recycling and disposal options<br>o E-waste disposal regulations<br>o Challenges in e-waste disposal<br>o Strategies for reuse, recycling, and disposal<br><br>• Emerging Trends in Green IT (09 hrs.)<br>o Green Cloud Computing<br>o Sustainable AI and machine learning<br>o Renewable energy integration in IT systems<br>o Internet of Things (IoT) for sustainability |
| **Practical** | • Practical 1: Field visit to an IT organization and prepare a report of Application of Green IT concept in that organization. Should consider the following factors when preparing the report:<br>o If no Green IT concept is used at present, should propose a Green IT system to them<br>o If they have a Green IT policy already, report should include proposals to improve their Green IT policy.<br>• Practical 2: Design an e-waste recycling plan for an IT organization<br>• Practical 3: Develop a Green IT strategy document for an IT organization<br>• Practical 4: Analysis of a real-world case studies of successful and unsuccessful (may be one each) Green IT initiatives |
| **Resources** | • Computer<br>• Internet Facility |

<table>
<tr><td></td><td colspan="3">• Relevant Software</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• GREEN COMPUTING AND ITS APPLICATIONS, SANJAY KUMAR, ROHIT RAJA, ALOK KUMAR SINGH KUSHWAHA, SAURABH KUMAR, RAJ KUMAR PATRA, ISBN: 978-1-68507-363-3<br>• Green Computing Tools and Techniques for Saving Energy, Money, and Resources, Bud E. Smith, [No ISBN]<br>• Green Home Computing for Dummies, Woody Leonhard and Katherine Murray, ISBN: 978-0-470-46745-9</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">• Lectures, Presentations, Demonstration, Practical, Discussions using the material</td></tr>
<tr><td rowspan="7">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="5">Continuous Assessment</td><td>Practical 01</td><td>10%</td></tr>
<tr><td>Practical 02</td><td>10&</td></tr>
<tr><td>Practical 03</td><td>10%</td></tr>
<tr><td>Practical 04</td><td>10%</td></tr>
<tr><td>Make two presentations based on two of the selected above practical</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -2 hour</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100 hrs. (48T + 30P + 22S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Advanced Techniques in Graphics Designing for Multimedia Content** |
| **Module Code** | K72T001M22/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U17, K72T001U18, K72T001U19 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>Create complex and professional graphics for multimedia content that make high visual communication with audience. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Produce sophisticated and professional graphic design projects across various media<br>2. Use advanced design tools and techniques to create innovative and compelling visuals<br>3. Explain the ethical, legal, and cultural dimensions of graphic design<br>4. Build a professional portfolio that demonstrates the design capabilities and creativity |
| **Learning Content / Topics** | • Graphic Design Theory - (12hrs.)<br>o Advanced color theory in design<br>o Typography for advanced graphic designers<br>o Visual hierarchy and composition techniques<br>o Psychological impact of design choices<br>o Cultural and societal implications of design<br><br>• Branding and Identity Design - (12hrs.)<br>o Principles of brand identity and brand development<br>o Logo design strategies and trends<br>o Brand guidelines and visual style guides<br>o Typography and color in branding<br><br>• Advanced Layout and Composition - (18hrs.)<br>o Advanced grid systems and alignment<br>o Multi-page document design (e.g. magazines, books, catalogs)<br>o Web and mobile layout considerations<br>o Responsive and adaptive design principles<br>o Working with templates for various platforms<br>o Designing for multiple formats (e.g. print, digital, social media, etc.) |

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| | • Digital and Interactive Design - (18hrs.)<br>  ○ Web design principles<br>  ○ Creating interactive prototypes<br>  ○ User experience (UX) and user interface (UI) design for websites and apps<br>  ○ Mobile app design principles and best practices<br><br>• Visual Communication - (06hrs.)<br>  ○ Data visualization for reports and presentations<br>  ○ Photography for graphic design<br>  ○ Designing for social media and digital campaigns<br><br>• Portfolio Development - (06hrs.)<br>  ○ Building a comprehensive and visually cohesive portfolio<br>  ○ Presentation and communication of design solutions<br>  ○ Personal branding and self-promotion techniques<br>  ○ Creating an online portfolio and leveraging social media<br><br>• Graphic Design Trends and Innovation - (09hrs.)<br>  ○ Emerging trends in graphic design (e.g. flat design, minimalism, etc.)<br>  ○ The role of artificial intelligence and automation in design<br>  ○ Future trends in user interfaces and experiences<br>  ○ Cross-cultural design considerations in a globalized world<br><br>• Ethical and Legal Aspects of Graphic Design - (09hrs.)<br>  ○ Copyright laws and intellectual property rights<br>  ○ Ethical considerations in design<br>  ○ Understanding contracts and agreements in design<br>  ○ Handling confidential client information<br>  ○ Managing disputes and feedback<br>  ○ Accessibility and inclusive design principles |
| **Practical** | • Practical 01: Create a report on Ethical use of Graphic designs in multimedia content<br>• Practical 02: Create an online portfolio<br>• Practical 03: Create visually appealing graphic design for a given scenario<br>• Practical 04: Design and develop user interface for a multimedia product or web page |
| **Resources** | • Computer<br>• Internet Facility |

<table>
<tr><td></td><td colspan="3"><ul><li>Relevant Software</li></ul></td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>Multimedia: Making It Work Ninth Edition, Tay Vaughan, ISBN: 978-0-07-183288-5</li></ul></td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3"><ul><li>Lectures</li><li>Presentations</li><li>Demonstration</li><li>Practical</li><li>Discussions using the material</li><li>Field visit</li></ul></td></tr>
<tr><td rowspan="6">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="4">Continuous Assessment</td><td>Practical 01</td><td>10%</td></tr>
<tr><td>Practical 02</td><td>10%</td></tr>
<tr><td>Practical 03</td><td>10%</td></tr>
<tr><td>Practical 04</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (2 hour)</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">200hrs. (90T +80 P + 30S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Design and develop audio, video, motion graphics and amination for multimedia content** |
| **Module Code** | K72T003M23/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U16 |
| **Pre-Requisites** | K72T001M22 |
| **Module Aim/s** | **To enable the students to:**<br>Equip learners with skills in audio and video production, motion graphics, and animation, enabling them to create high quality multimedia content for various platforms. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain fundamentals of multimedia content and their applications<br>2. Explain the role of audio, video, motion graphics, and animation in modern media including visual projection on LED walls<br>3. Design and produce professional-quality audio for multimedia content<br>4. Produce and edit professional-quality video for various formats<br>5. Explain the principles of animation and apply them in motion graphics |
| **Learning Content / Topics** | • Multimedia Production/Content (12hrs.)<br>o Overview of multimedia content creation (audio, video, motion graphics, and animation)<br>o Copywriting techniques<br>o Applications and uses in various areas (e.g. advertising, social media, search engines, film, gaming, education, digital marketing, etc.)<br>o Digital Formats (Audio, Video, Animation, etc.)<br>o Industry-standard tools and software for multimedia production<br>o Trends and innovations in multimedia design<br><br>• Audio Design and Production - (12hrs.)<br>o Audio recording techniques and equipment (microphones, mixers, software)<br>o Editing and mixing using software<br>o Sound design for multimedia (background, sound effects, voiceovers)<br>o Synchronizing audio with visual elements<br>o Exporting and optimizing audio for different platforms |

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| | • Video Production and Editing - (18hrs.)<br>  ○ Video Pre-Production<br>  ▪ Scripting and storyboarding<br>  ▪ Camera techniques and shot planning<br>  ○ Video Production<br>  ▪ Lighting setups and techniques<br>  ▪ Camera operation<br>  ○ Video Editing<br>  ▪ Non-linear editing techniques<br>  ▪ Transitions, effects, and color grading<br>  ○ Post-Production<br>  ▪ Motion tracking and stabilization<br>  ▪ Audio-video synchronization<br><br>• Motion Graphics Design - (15hrs.)<br>  ○ Introduction to Motion Graphics<br>  ▪ Principles of motion design<br>  ▪ Storyboarding for motion graphics<br>  ○ Creating Motion Graphics<br>  ▪ Keyframing and animation<br>  ▪ 3D motion graphics<br>  ▪ Integrating typography with motion<br>  ○ Exporting and Optimization<br>  ▪ Formats for web and broadcast<br>  ▪ Compression techniques<br><br>• Animation (2D, 3D and Character) - (12hrs.)<br>  ○ Animation Principles (e.g. squash and stretch, anticipation, timing, etc.)<br>  ○ Animation Techniques (e.g. keyframe, procedural, motion capture, etc.)<br>  ○ Rendering techniques<br>  ○ Optimizing animations for platforms (e.g. video, web, social media, etc.)<br><br>• Integration of Audio, Video, Motion Graphics and Animations - (03hrs.)<br>• Visual projection, mapping and Playback - (18 hrs.)<br>  ○ Visual projection Devices<br>  ○ Setup, configure and commissioning visual projection devices<br>  ○ Visual projection Software (E.g.: Vmix, Resolume Areena) |

<table>
<tr><td>Practical</td><td colspan="3">• Practical 01: Create an audio track for a media project/given scenario<br>• Practical 02: Produce a short video project for a given scenario<br>• Practical 03: Create motion graphics for a selected/given media<br>• Practical 04: Integrate motion graphics into video projects<br>• Practical 05: Develop a short-animated sequence for a given scenario<br>• Practical 06: Prepare a report by attending to an event that uses visual projection<br>• Practical 07: Develop a comprehensive multimedia project from concept to completion. The project should include audio, video, motion graphics, and animation in it and present the multimedia project</td></tr>
<tr><td>Resources</td><td colspan="3">• Computer<br>• Internet Facility<br>• Relevant Software</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Design for motion: fundamentals and techniques of motion design, Shaw, Austin; Shaw, Danielle, ISBN: 9781138318649</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">• Lectures<br>• Presentations<br>• Demonstration<br>• Practical<br>• Discussions using the material</td></tr>
<tr><td rowspan="7">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="5">Continuous Assessment</td><td>Practical 01</td><td>10%</td></tr>
<tr><td>Practical 02</td><td>10%</td></tr>
<tr><td>Practical 03</td><td>10%</td></tr>
<tr><td>Practical 05</td><td>10%</td></tr>
<tr><td>Practical 07</td><td>40%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (01 hour)</td><td>20%</td></tr>
<tr><td>Duration</td><td colspan="3">200hrs. (90T + 80P + 30S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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|---|---|
| **Module Title** | **Manage cloud resources** |
| **Module Code** | K72T001M24/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U22 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>Equip learners with the knowledge and skills necessary to effectively manage cloud resources across various platforms (e.g., AWS, Azure, GCP), ensuring optimal performance, security, and cost-efficiency. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain cloud computing models and architectures<br>2. Manage cloud resources effectively, including provisioning and scaling<br>3. Implement cost management strategies for cloud services<br>**4.** Ensure security and compliance in cloud environments |
| **Learning Content / Topics** | • Cloud Computing Fundamentals (06 hrs.)<br>o Overview of cloud computing<br>o Benefits and challenges of cloud adoption<br>o Cloud computing architecture and layers<br>o Cloud service models (IaaS, PaaS, SaaS)<br>o Cloud Deployment Models (Public, Private, Hybrid, Community)<br>o Key Cloud Providers (AWS, Azure, GCP, etc.)<br>• Core Cloud Services (06 hrs.)<br>o Computing Services (Virtual Machines, Containers, Serverless Computing)<br>o Storage Services (Object Storage, Block Storage, File Storage)<br>o Networking Services (Virtual Networks, Load Balancing, VPNs)<br>o Database Services (Relational Databases, NoSQL Databases, Data Warehouses**)**<br>o Security services<br>• Managing Cloud Infrastructure (06 hrs.)<br>o Setting up virtual machines and containers<br>o Configuring cloud storage solutions<br>o Networking in the cloud: (VPCs, subnets, and security groups, etc.)<br>• Monitoring and Optimization (06 hrs.)<br>o Using cloud-native monitoring tools (e.g. AWS CloudWatch, Azure Monitor)<br>o Analyzing performance metrics and reports |

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| | ○ Automating scaling policies (e.g. autoscaling groups)<br>○ Optimizing resource allocation and costs.<br><br>• Cloud Security Management (06 hrs.)<br>○ Cloud security fundamentals<br>○ Identity and Access Management (IAM)<br>○ Data Security and Encryption<br>○ Network Security (Firewalls, VPNs, Intrusion Detection Systems)<br>○ Best practices for securing cloud environments<br>○ Compliance frameworks (e.g., GDPR, ISO 27001)<br><br>• Cloud Native Technologies (06 hrs.)<br>○ Containers and Container Orchestration (Docker, Kubernetes)<br>○ Microservices Architecture<br>○ Serverless Computing (Lambda, Functions)<br>○ Cloud-Native Applications and Development<br><br>• Cloud Cost Management (06hrs)<br>○ Understanding cloud pricing models<br>○ Tools for cost analysis and reporting<br>○ Budgeting and forecasting cloud expenditures<br>○ Techniques for reducing cloud costs<br><br>• Hybrid Cloud Management (06 hrs.)<br>○ Integrating on-premise and cloud resources<br>○ Managing data flow between environments<br>○ Challenges in hybrid cloud implementation<br><br>• Cloud Solution Evaluation and Selection (06 hrs.)<br>○ Assessing the requirements of a business for cloud migration<br>○ Comparing cloud providers (AWS vs. Azure vs. Google Cloud)<br>○ Developing a migration plan<br><br>• Emerging Trends in Cloud Computing (06 hrs.)<br>○ Overview of current trends (AI, IoT, serverless computing)<br>○ Future directions in cloud technology<br>○ Impact of emerging technologies on cloud resource management |
| **Practical** | Practical 1: Deploy a scalable web application on a cloud platform<br>Practical 2: Design a disaster recovery plan for a given scenario<br>Practical 3: Monitor and optimize a cloud-based infrastructure for performance and cost efficiency<br>Practical 4: Implementing security best practices in cloud environments |

<table>
<tr><td>Resources</td><td colspan="3"><ul><li>Multimedia, Whiteboard</li><li>Smart board</li><li>Computer lab with internet facility</li></ul></td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>The complete Cloud Computing Manual, BDM publications,</li><li>The Cloud Computing Book, DOUGLAS E. COMER, ISBN: 9781003147503</li></ul></td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">Lectures, Presentations, Discussions using the material and product samples</td></tr>
<tr><td rowspan="7">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="5">Continuous Assessment</td><td>Practical 01</td><td>15%</td></tr>
<tr><td>Practical 02</td><td>15%</td></tr>
<tr><td>Practical 03</td><td>15%</td></tr>
<tr><td>Practical 04</td><td>15%</td></tr>
<tr><td>Make a presentation based on one of the selected above practical</td><td>10%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -01 hour</td><td>30%</td></tr>
<tr><td>Duration</td><td colspan="3">75hrs. (60T + 30P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Enterprise network setup and configuration** |
| **Module Code** | K72T001M25/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U20 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Develop skills in setting up, configuring and troubleshooting hardware, software and security settings in an enterprise network. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain networking concepts, including protocols, models, and architectures<br>2. Install and configure network devices<br>3. Monitor and troubleshoot network issues<br>**4.** Implement security measures |
| **Learning Content / Topics** | • Networking Concepts (12 hrs.)<br>o Networking Fundamentals<br>o Network Types (LAN, WAN, MAN, WLAN, etc.)<br>o Overview of hardware components (switches, routers, cables, access points, etc.)<br>o Networking protocols (TCP/IP, OSI model)<br>o Understanding IP addressing (IPv4 and IPv6)<br>o Network topologies and their applications<br>o VLANs, VPNs, and network segmentation<br>o Setting up sharing of resources (e.g. file, print, etc.)<br>o Network security fundamentals<br><br>• Enterprise Network Solutions (06 hrs.)<br>o Importance of enterprise networks<br>o Components of enterprise networks (routers, switches, firewalls, etc.)<br><br>• Network Design Concepts (06 hrs.)<br>o Network design principles<br>o Network diagrams and documentation<br><br>• Network Setup and Configuration (18 hrs.)<br>o Network devices and their configurations<br>o Configuring servers (DNS, DHCP, etc.)<br>o Basic router and switch configuration (CLI basics)<br>o Advanced router and switch configuration (VLANs, trunking)<br>o Implementing routing protocols (RIP, OSPF, EIGRP)<br>o Network Automation and Scripting (Python, Ansible) |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Network segmentation (subnetting, Supernetting and CIDR notation)</li>
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<ul>
<li>Network Monitoring and Troubleshooting (06 hrs.)
<ul>
<li>Tools and techniques for network monitoring (Ping, Traceroute, SNMP, NetFlow, etc.)</li>
<li>Common troubleshooting methodologies and tools</li>
<li>Troubleshooting wireless connectivity issues</li>
</ul></li>
<li>Security Measures (06 hrs.)
<ul>
<li>Firewalls and intrusion detection/prevention systems</li>
<li>Security protocols for wireless networks (WPA, WPA2, WPA3)</li>
<li>Best practices for securing enterprise networks</li>
</ul></li>
<li>Cloud Networking and Virtualization (12 hrs.)
<ul>
<li>Cloud-based network solutions (e.g. Azure, AWS, etc.)</li>
<li>Configuring and managing virtual networks (SDN, NFV)</li>
</ul></li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">
<ul>
<li>Practical 01- Configure a basic router and switch for network communication</li>
<li>Practical 02 - Set up and secure a VPN connection</li>
<li>Practical 03- Prepare a report on network performance analysis and recommendations (including network designs, configurations, and security measures)</li>
<li>Practical 04 - A simulated project that involves designing, implementing, and configuring an enterprise network solution, including documentation and presentation to stakeholders</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>e Resource/ Internal resources (Routers, switches, connecters, servers- DHCP, DNS, Internet connectivity and other related components)</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>Architecture of Network Systems, Dimitrios Serpanos, Tilman Wolf, ISBN: 9780123744944</li>
<li>The Art of Network Architecture: Business-Driven Design, White Russ, Donohue Denise, ISBN: ISBN: 978-0-13-325920-9</li>
</ul>
</td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">
<ul>
<li>Lectures</li>
<li>Presentations</li>
<li>Discussions using the material and product samples</li>
</ul>
</td></tr>
<tr><td rowspan="6">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="4">Continuous Assessment</td><td>Practical 01</td><td>10%</td></tr>
<tr><td>Practical 02</td><td>10%</td></tr>
<tr><td>Practical 03</td><td>10%</td></tr>
<tr><td>Practical 04</td><td>30%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -2 hour</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100 hrs. (66T + 22P + 12S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Maintain Network solution** |
| **Module Code** | K72T001M26/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U21 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>Equips learners with the knowledge and skills necessary to effectively maintain and optimize network infrastructures, ensuring high availability, performance, and security. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain the principles of network design and architecture<br>2. Implement maintenance strategies for network solutions<br>3. Troubleshoot and resolve network issues effectively<br>4. Monitor network performance and optimize resources |
| **Learning Content / Topics** | • Network Maintenance (09 hrs.)<br>o Introduction to network maintenance and its importance<br>o Types of network solutions (LAN, WAN, WLAN, etc.)<br>o Standards and protocols (IEEE, ISO, TCP/IP, UDP, and OSI model, etc.)<br>o Overview of preventive, corrective, and predictive maintenance<br>• Network Hardware and Software (06 hrs.)<br>o Understanding network devices (e.g. Routers, switches, repeaters, firewalls, access points, servers, etc.)<br>o Firmware and software updates for networking equipment<br>o Cable types and termination (twisted pair, fiber optics, and coaxial)<br>o Managing hardware inventory and lifecycle<br>• Network Performance Monitoring, Diagnostics and Optimization (06 hrs)<br>o Tools and techniques for network monitoring (e.g., Wireshark, Nagios, SolarWinds)<br>o Network Performance Metrics (latency, jitter, throughput, packet loss)<br>o Network Monitoring Tools (e.g. SNMP, NetFlow, RMON, OTDR)<br>o Identifying and diagnosing network issues: Loopbacks, signal interference, and bottlenecks<br>o Performance Analysis and Troubleshooting<br>o Network Optimization Techniques (QoS, traffic shaping, bandwidth management)<br>o Logging and analyzing network events and incidents<br>o Redundancy planning and performing<br>• Preventive Maintenance (03 hrs.)<br>o Establishing maintenance schedules for network devices<br>o Cleaning and inspecting physical components |

| | |
|---|---|
| | ○ Regular backups of network configurations and data<br>○ Ensuring compliance with network standards and policies<br><br>• Network Security Maintenance (06 hrs.)<br>○ Network Security Threats and Vulnerabilities<br>○ Intrusion Detection and Prevention Systems (IDS/IPS)<br>○ Firewall Configuration and Management<br>○ Configuring VPNs<br>○ Implementing endpoint security solutions and anti-malware tools<br>○ Vulnerability Scanning and Penetration Testing<br>○ Security Information and Event Management (SIEM)<br><br>• Network Troubleshooting and Remediation (06 hrs.)<br>○ Common Network Problems and Troubleshooting Techniques<br>○ Network Troubleshooting Tools (ping, traceroute, Wireshark)<br>○ Fault Isolation and Diagnosis<br>○ Resolving common issues: DNS failures, IP conflicts, and connectivity drops<br>○ Network Restoration and Recovery Procedures<br>○ Incident Response and Management<br><br>• Backup and Disaster Recovery (06 hrs.)<br>○ Planning and implementing network backup strategies<br>○ Disaster recovery plans for network failures<br>▪ Disaster Recovery Drill<br>○ Restoring network services from backup configurations<br>○ Identifying and mitigating single points of failure (SPOFs)<br><br>• Performance Optimization (12 hrs.)<br>○ Techniques to optimize bandwidth usage and reduce latency<br>○ Quality of Service (QoS) configuration for critical applications<br>○ Upgrading network devices to support higher capacity<br>○ Using load balancers for efficient traffic distribution<br><br>• Emerging Trends in Network Maintenance (06 hrs.)<br>○ Overview of current trends (SDN, NFV, IoT)<br>○ Automation of maintenance tasks using tools and scripts.<br>○ The role of artificial intelligence (AI) in predictive network maintenance.<br>○ Cloud-based network management solutions.<br>○ Impact of emerging technologies on network maintenance<br>○ Future directions in network solutions |
| **Practical** | • Practical 1: Setting up and maintaining a network |

<table>
<tr><td></td><td colspan="3"><ul><li>Practical 2: Develop and implement a maintenance plan for a given network scenario</li><li>Practical 3: Simulate network failures and practice recovery procedures</li><li>Practical 4: Configure and secure a wireless network</li></ul></td></tr>
<tr><td>Resources</td><td colspan="3"><ul><li>e Resource/ Internal resources (Routers, switches, connecters, servers- DHCP, DNS, Internet connectivity, Access point and other related components)</li></ul></td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>Practical Packet Analysis: Using Wireshark to Solve Real-World Network Problem, Chris Sanders, ISBN: 9781593278021</li><li>Computer Network Solution 4th Ed., Andrew S. Tannenbaum, David J. Wetherall, ISBN: 0-13-046002-8</li></ul></td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3">Lectures, Presentations, Discussions using the material and product samples</td></tr>
<tr><td rowspan="7">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="5">Continuous Assessment</td><td>Practical 01</td><td>15%</td></tr>
<tr><td>Practical 02</td><td>15%</td></tr>
<tr><td>Practical 03</td><td>15%</td></tr>
<tr><td>Practical 04</td><td>15%</td></tr>
<tr><td>Make a presentation based on one of the selected above practical</td><td>10%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam (2 hour)</td><td>30%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T +30P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Operating Systems and Application Security** |
| **Module Code** | K72T001M27/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U25 |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>To explore the security features and vulnerabilities of operating systems and applications, equipping learners to safeguard systems against threats, and in-depth understanding of operating system security principles, application security mechanisms, and their role in securing IT environments. |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Understand the role of operating systems in system security<br>2. Apply secure configurations to operating systems and applications<br>3. Identify and address common application vulnerabilities<br>4. Conduct security assessments and penetration testing of operating systems and applications |
| **Learning Content / Topics** | • Operating Systems Security (12 hrs.)<br>o Operating System Architecture and Security Features<br>o Operating system vulnerabilities and exploits<br>o User authentication and authorization<br>o Access control mechanisms<br><br>• Securing Operating Systems (12 hrs.)<br>o Secure Configuration of the operating system (e.g. Windows, Linux etc.)<br>o Implementing Group Policies and Security Policies<br>o Patch Management and Update Practices<br>o Hardening Techniques for Servers and Workstations<br><br>• Application Security Fundamentals (6 hrs.)<br>o Types of Applications (e.g. Desktop, Web, Mobile, and Cloud-Based)<br>o Common Application Vulnerabilities<br>o Secure Development Life Cycle (SDLC)<br>o Role of Application Security in Cyber Defense<br><br>• Vulnerability Assessment and Mitigation (12 hrs.)<br>o Tools for Vulnerability Assessment (Nessus, OpenVAS, Nmap, Burp Suite, etc.)<br>o Secure Input Validation and Output Encoding<br>o Implementing Authentication and Authorization Mechanisms |

| | |
|---|---|
| | ○ Preventing Injection Attacks and Cross-Site Scripting (XSS)<br><br>• Encryption and Secure Communication (6 hrs.)<br>○ Encryption Standards (e.g. AES, RSA, TLS, etc.)<br>○ Using Secure Protocols for Data Transmission (HTTPS, SFTP, etc.)<br>○ Application-Level Encryption Practices<br>○ Digital Signatures and Certificates in Application Security<br><br>• Legal and Compliance Aspects (12 hrs.)<br>○ Compliance Requirements (e.g. PCI DSS, GDPR, HIPAA, etc.)<br>○ Regulatory Standards for Secure Applications<br>○ Logging, Monitoring, and Reporting in Secure Applications |
| **Practical** | • Practical 1: User Account Management - Create and manage user accounts with different levels of privileges (administrator, standard user, guest). Implement strong password policies and enforce regular password changes.<br>• Practical 2: System Hardening -Disable unnecessary services, close unused ports, and configure firewall rules to restrict network access. Install and configure security updates and patches.<br>• Practical 3: Security Auditing -Monitor system logs for suspicious activity, such as failed login attempts, unauthorized access, and malware infections.<br>• Practical 4: Malware Analysis -Analyze the behavior of a malware sample in a sandbox environment.<br>• Practical 5: Backup and Recovery -Perform a backup and recovery operation |
| **Resources** | • Computer<br>• Internet Facility<br>• Relevant Software |
| **Prescribed Texts & / or References** | • Computer and Information Security Handbook - 3rd Edition, John R. Vacca<br>• The Complete Cyber Security Course, Volume 1 Hackers Exposed, Nathan House<br>• Principles of Information Security - 5th Edition, Whitman, Michael Mattord, Herbert<br>• Foundations of Information Security A Straightforward Introduction, Jason Andress<br>• Cybersecurity for Dummies, Joseph Steinberg<br>• Network Security for Dummies, 1st Edition, Chey Cobb |
| **Teaching/Learning Activities** | • Lectures<br>• Presentations<br>• Demonstration<br>• Practical<br>• Discussions using the material |

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| **Assessment & Weighting** | **Type** | **Topic / Activity/Learning outcomes** | **Module Weighting** |
| | Continuous Assessment | Practical 01 | 10% |
| | | Practical 02 | 10% |
| | | Practical 03 | 10% |
| | | Practical 04 | 10% |
| | | Practical 05 | 10% |
| | | Make a presentation based on one of the selected above practical | 10% |
| | Summative Assessment | Written test (2 hour) | 40% |
| **Duration** | **100hrs. (60T + 30P + 10S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

# MODULE DESCRIPTORS
# Information and Communication Technology
# NVQ Level 06
## Semester I
## Employability modules

| | |
|---|---|
| **Module Title** | **Problem Solving and Decision Making** |
| **Module Code** | EMTM04/Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | EMTU04/Level 06 |
| **Pre-Requisites** | None |
| **Module Aims** | **To enable the students to**<br>▪ Understand the importance of following a systematic approach to identifying and analyzing problem situation/s<br>▪ Acquire the skills required to compare different solutions to decide the most suitable.<br>▪ Apply standard techniques of problem solving and decision making to solve problems and make decisions at workplace.<br>▪ |
| **Learning Outcomes** | **The student shall be able to**<br>▪ Analyze a situation and accurately identify a problem<br>▪ Assess the impact of a problem<br>▪ Apply systematic procedure and process of solving a problem<br>▪ Follow a standard approach to foresee likely problems<br>▪ Follow an acceptable decision-making process |
| **Learning Content / Topics** | ▪ Problem solving models<br>▪ Problem solving techniques<br>▪ Decision making models<br>▪ Decision making process- (Reference to two commonly used decision models)<br>▪ Decision making styles – (Participatory approaches may be given priority)<br>▪ Creative decision making - (Particular reference to six stages in Creative decision-making process)<br>▪ Six stages in Creative decision-making process<br>▪ Recognition<br>▪ Fact finding<br>▪ Problem finding<br>▪ Idea finding<br>▪ Solution finding<br>▪ Acceptance finding<br>▪ Vroom and Yetter's tool for decision making - "Decision tree"<br>▪ Brainstorming technique<br>▪ Nominal group technique<br>▪ Front end analysis |

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| **Resources** | Occupational Tools, Instruments, Equipment, material<br>• Documents on Company policies and procedures<br>• Documents on Industry/Workplace codes of Practice/Codes of ethics<br>• Documents on Manufacturer' s specifications and product specifications<br>• Model of Vroom and Yetter's tool for decision making - "Decision tree"<br>• Calculator<br>Instructional Tools, Instruments, Equipment, material<br>• Multimedia projector<br>• Screen<br>• Computer<br>• Flash cards<br>• Flip charts<br>• Permanent and white board markers<br>• Overhead projector<br>• Transparencies<br>• Color Printer and scanner<br>• Illustrations of "Problem solving models"<br>• Illustrations of "Problem solving techniques"<br>• Illustrations of "Decision making models" |
| **Recommended Texts & / or References** | |
| **Teaching Learning Activities** | ▪ (Small group activity -1) -Provide relevant case studies to trainees and make them work in small groups to identify problems and the impact of the problems on one's immediate area of responsibilities<br>▪ (Small group activity -2) -Get trainees work in small groups to generate ideas using divergent and convergent approaches to create solutions to problems identified in activity -1<br>▪ (Small group activity -3) Get trainees work in small groups to select and communicate the most appropriate solution from among those generated in activity -2<br>▪ Explain<br>• Problem solving models<br>• Problem solving techniques<br>• Decision making models<br>• Decision making process- (Reference to two commonly used decision models)<br>• Decision making styles – (Participatory approaches may be given priority) |

<table>
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- Creative decision making -

Provide a real problem situation (case study) to follow the Six stages in "Creative decision making" process to make appropriate decisions to problems identified in the above case study

- Describe Vroom and Yetter's tool for decision making - "Decision tree"
  - Brainstorming technique
  - Nominal group technique
  - Front end analysis
- Provide examples from real workplace situations to make trainees apply principles learnt in each of the above techniques

</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity</td><td>Weighting</td></tr>
<tr><td rowspan="2">Formative assessment</td><td>Skills in problem solving / decision making at workplace during industry training</td><td>50%</td></tr>
<tr><td>Knowledge of various topics on problem solving / decision making- oral questioning technique during class room presentations</td><td>10%</td></tr>
<tr><td rowspan="2">Summative assessment</td><td>Knowledge of various topics on problem solving / decision making - Multiple Choice Test Items/ Matching and Completion test items and structured essay type questions</td><td>30%</td></tr>
<tr><td></td><td>knowledge in applications of problem solving / decision making - Viva voce</td><td>10%</td></tr>
<tr><td>Duration</td><td colspan="3">50 hours (30T + 15P + 05S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

# MODULE DESCRIPTORS INFORMATION AND COMMUNICATION TECHNOLOGY NVQ Level 06

## Semester 2

## Technical modules

| | |
|---|---|
| **Module Title** | **DevOps and Site Reliability Engineering (SRE)** |
| **Module Code** | K72T001M28/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U12 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Understand the development of collaborative culture within the teams to enhance the integration and delivery process |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>• Explain the methodologies use in DevOps and SRE<br>• Apply DevOps tools to enhance software delivery process, ensure system reliability, and implement automation for operational tasks |
| **Learning Content / Topics** | ● Introduction to DevOps and SRE – (3 hrs.)<br>○ Key Concepts and Differences<br>○ Collaboration, automation, and reliability in software development<br>● DevOps Fundamentals – (12 hrs.)<br>○ DevOps life cycle<br>○ Version Control with Git and Remote<br>○ Automation in Development and Deployment Pipelines<br>○ Infrastructure as Code (IaC)<br>○ Cloud Automation and Auto-Scaling (AWS, GCP, Azure, etc.)<br>● SRE Principles – (15 hrs.)<br>○ Service Level Objectives (SLOs)<br>○ Reducing Toil through Automation<br>○ Monitoring and Observability<br>■ Key Metrics<br>- Service Level Indicators (SLIs)<br>- Latency<br>- Throughput<br>- Error Rates<br>■ Performance Monitoring Tools<br>■ Centralized Logging<br>■ Service Level Agreements (SLAs)<br>● Cloud and Containerization – (12 hrs.)<br>○ Cloud Platforms (AWS, GCP, Azure, etc.) Overview |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Containerization with Docker</li>
<li>Kubernetes for Container Orchestration</li>
</ul>
<ul>
<li>Automation and CI/CD Pipelines – (6 hrs.)
<ul>
<li>Automating Builds and Deployments (Jenkins, GitLab CI, etc.)</li>
</ul>
</li>
<li>Performance and Capacity Planning – (3 hrs.)
<ul>
<li>Scaling Strategies and Cloud Auto-scaling</li>
<li>Load Balancing and Performance Optimization</li>
<li>Capacity Planning and Predictive Scaling</li>
</ul>
</li>
<li>Security in DevOps – (15 hrs.)
<ul>
<li>Integrating Security in CI/CD Pipelines (DevSecOps)</li>
<li>Identity and Access Management (IAM)</li>
<li>Securing Cloud Infrastructure</li>
</ul>
</li>
<li>DevOps setup and configuration documents – (6 hrs.)</li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">
<ul>
<li>Assignment 01: Prepare a report to explain the experience gained during application of DevOps tools in development process</li>
<li>Assignment 02: Prepare a report of the DevOps configuration and setup details of the project developed</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Computer</li>
<li>Internet Facility</li>
<li>Relevant Software</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>The DevOps handbook. Second edition, Gene Kim & Jez Humble & Patrick Debois & John Willis & Nicole Forsgren, ISBN: 9781950508433</li>
<li>Modern DevOps Practices - Second Edition: Implement, secure, and manage applications on the public cloud by leveraging cutting-edge tools, Gaurav Agarwal,</li>
</ul>
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
<ul>
<li>Lectures</li>
<li>Presentations</li>
<li>Demonstration</li>
<li>Practical</li>
<li>Discussions using the material</li>
</ul>
</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Assignment 01</td><td>20%</td></tr>
<tr><td>Assignment 02</td><td>40%</td></tr>
<tr><td>Summative Assessment</td><td>Presentation followed by VIVA Voce for the prepared report</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (72T + 20P + 08S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Software Development final Project** |
| **Module Code** | K72T001M29/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U10, K72T001U11, K72T001U12, K72T001U13, K72T001U14, K72T001U15 |
| **Pre-Requisites** | Successfully completion previous modules in semester I and II |
| **Module Aim/s** | **To enable the students to:**<br>Develop a computer software for an organization to automate their operations. The project is providing an opportunity for students to apply concepts, methods and skills acquired during the course to solve a real-life business or organizational situation. The project will be carried out in individual basis |
| **Learning Outcome/s** | **At the end of this module the student shall be able to**<br>1. Identify the software requirements<br>2. Propose the solutions for identified requirements<br>3. Develop the required software application<br>4. Test and deploy the software application<br>5. Prepare software development project report |
| **Project components and related instructions** | •Project proposal- (10Hrs.)<br>o Project requirements<br>o Proposed solutions<br>o Cost and time schedule<br>•First draft report –(06Hrs.)<br>•Interim report – (06 Hrs.)<br>o Should make a progress presentation<br>•Final system and report – (10 Hrs.)<br>o Make a demonstration on developed software application<br>o Viva voce<br>o Final report<br>•Report content-(18 Hrs.)<br>o Software Requirement Specifications (SRS)<br>o Required diagrams (E.g. ER, Use case, class, activity, etc.)<br>o UI diagrams<br>o Deployment plan (DevOps configuration)<br>o User Acceptance Test procedure (UAT procedure)<br>o Future enhancement |

<table>
<tr><td>Prescribed Texts & / or References</td><td>

- Research Methods for business – Dr.Uma Sekaran and Dr.Roger Bougie
- Writing Research Proposals – Prof. Jayadeva Uyangoda

</td></tr>
<tr><td>Teaching/ Learning Activities</td><td>

- Present project instructions
- search, investigate independently of a practical problem
- report writing
- Analyze the problem and interpret results
- Communicate effectively the status of an activity in technical and logical manner by presenting to a technical audience

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<table>
<tr><th>Marks Breakdown</th><th>Marks (%)</th></tr>
<tr><td colspan="2"></td></tr>
<tr><td>Project proposal with presentation</td><td>10</td></tr>
<tr><td>First draft report</td><td>10</td></tr>
<tr><td>Interim report with presentation</td><td>20</td></tr>
<tr><td>System demonstration with viva</td><td>30</td></tr>
<tr><td>Final Report</td><td>30</td></tr>
<tr><td>Total</td><td>100</td></tr>
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</td></tr>
<tr><td>Duration</td><td>200.hrs. (50T + 0P + 150S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
</table>

| | |
|---|---|
| **Module Title** | **Artificial Intelligence (AI)** |
| **Module Code** | K72T001M30/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | All Units |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>This module equips learners with the foundational knowledge of Artificial Intelligence (AI) techniques and tools that could be uses to design, develop and implement their projects easily. It covers key AI concepts, their applications, AI tools and libraries. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Explain the differences between AI, machine learning, and deep learning<br>2. Explain the history and evolution of AI<br>3. Explain the ethical implications of AI technologies<br>4. Discuss the applications of AI technologies/tools in various fields |
| **Learning Content / Topics** | • Introduction to Artificial Intelligence (AI) – (21hrs.)<br>o Definitions of AI<br>o AI types (Narrow AI, General AI)<br>o Subfields of AI (Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision)<br>o History and evolution of AI<br>o AI Applications in various fields (e.g. Web design/development, Software development, etc.)<br>o Ethical considerations and societal impact of AI<br>• Machine Learning - (18hrs.)<br>o Supervised, unsupervised, and reinforcement learning<br>o Regression, classification, clustering algorithms<br>o Machine learning libraries (e.g. scikit-learn, TensorFlow.js, etc.)<br>o Model training, evaluation, and optimization<br>• Natural Language Processing (NLP) -(15hrs.)<br>o Text preprocessing and analysis<br>o NLP concepts (e.g. Sentiment analysis, text classification, topic modeling, etc.)<br>o NLP libraries (e.g. NLTK, spaCy, etc.)<br>o Building chatbots using NLP<br>• Computer Vision (Image and Video Processing) - (15hrs.)<br>o Image recognition and object detection algorithms/ techniques<br>o Image recognition, object detection, facial recognition<br>o Computer vision libraries (e.g. OpenCV.js, TensorFlow.js, etc.) |

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<tr><td></td><td colspan="3">
<ul>
<li>Image-based search, visual recognition features</li>
</ul>
<ul>
<li>Cloud AI Services - (03hrs.)
<ul>
<li>Overview of AI cloud platforms (e.g. AWS AI, Google Cloud AI, Azure AI, etc.)</li>
<li>Security and scalability of AI systems</li>
</ul>
</li>
<li>Future of AI? - (03hrs.)</li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">
<ul>
<li>Assignment 01 - Prepare a report on AI misuse and ethical dilemmas for given Case study/studies</li>
<li>Assignment 02 -Prepare a report on ethical considerations in AI for given Case study/studies</li>
</ul>
and<br>
Practical 01: perform one of the following tasks:
<ul>
<li>Develop a real-world AI-powered web application</li>
<li>Create and optimize advertising campaigns using AI tools</li>
<li>Design an enterprise network architecture using AI tools</li>
</ul>
</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Computer</li>
<li>Internet Facility</li>
<li>Relevant Software</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>AI for absolute beginners, Oliver Theobald, [No ISBN]</li>
<li>How to Teach AI, Rachelle Dené Poth, ISBN: 979-8-88837-035-3</li>
<li>Developing AI Applications An Introduction, Metin Karatas, ISBN 978-1-4932-2603-0</li>
</ul>
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
<ul>
<li>Lectures</li>
<li>Presentations</li>
<li>Demonstration</li>
<li>Practical</li>
<li>Discussions using the material</li>
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<tr><td rowspan="6">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="4">Continuous Assessment</td><td>Assignment 01</td><td>10%</td></tr>
<tr><td>Assignment 02</td><td>10%</td></tr>
<tr><td>Practical 02</td><td>30%</td></tr>
<tr><td>Make presentation based on above practical 02</td><td>10%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -2 hour</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">150hrs. (75T + 60P + 15S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Business Continuity Planning and Disaster Recovery** |
| **Module Code** | K72T001M31/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | All Units |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>Use the principles and practices of business continuity planning (BCP) and disaster recovery (DR), to ensure the organizational resilience during disruptions |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Explain the concepts and importance of Business Continuity Planning and Disaster Recovery<br>2. Recognize the importance of BCP and DR in organizational resilience<br>3. Conduct risk assessments and business impact analyses to identify critical systems and operations<br>4. Develop, test, and maintain business continuity and disaster recovery plans with mock/drill exercises<br>5. Explain the roles and responsibilities of key stakeholders during disruptions |
| **Learning Content / Topics** | • Business Continuity and Disaster Recovery (12 hrs.)<br>o Definitions and Objectives of Business Continuity and Disaster Recovery<br>o Importance of BCP and DR in Modern Organizations<br>o Key Concepts: Resilience, Recovery, Continuity<br>o Role of stakeholders in BCP and DR<br><br>• Business Impact Analysis (BIA) and Risk Assessment (12 hrs.)<br>o Purpose and Scope of a BIA<br>o Identifying Critical Business Functions and Dependencies<br>o Analyzing potential threats and impacts<br>o Estimating Downtime Impacts: Financial, Operational, Reputational<br>o Prioritizing Risks and Developing Mitigation Strategies<br><br>• Disaster Recovery Planning (DRP) (09 hrs.)<br>o Developing DR strategies and plans<br>o IT Disaster Recovery (Data Backup, Recovery Solutions, Tools, etc.)<br>o Incident response and crisis management<br>o Cloud-Based DR Solutions and Virtualization<br>o Testing, maintaining and communication of DR plans |

<table>
<tr><td></td><td colspan="3">
<ul>
<li>Business Continuity Planning (09 hrs.)
<ul>
<li>Components of a Business Continuity Plan.</li>
<li>Emergency Response, Communication, and Escalation Procedures.</li>
<li>Establishing Roles and Responsibilities in a BCP</li>
<li>Redundancy and Backup Strategies</li>
<li>Alternative Work Arrangements (e.g., Remote Work, Hot Sites, Cold Sites, etc.)</li>
<li>Vendor and Supply Chain Continuity Management</li>
</ul></li>
<li>Emerging Trends and Technologies (09 hrs.)
<ul>
<li>Cloud computing and its impact on BCP and DR</li>
<li>Cybersecurity and its role in BCP and DR</li>
<li>Pandemic planning and response</li>
<li>Business continuity in the age of globalization and digital transformation</li>
</ul></li>
<li>Training and Awareness Programs (09 hrs.)
<ul>
<li>Developing Training Programs for Key Stakeholders</li>
<li>Creating Awareness of Continuity and Recovery Procedures</li>
<li>Best Practices for Stakeholder Engagement</li>
</ul></li>
</ul>
</td></tr>
<tr><td>Practical</td><td colspan="3">Practical 1: Develop BCP and DR Plans<br>Practical 2: Simulate a Disaster Recovery Scenario (Mock/Drill session)<br>Practical 3: Prepare a Risk Assessment and BIA Report for a given scenario</td></tr>
<tr><td>Resources</td><td colspan="3">
<ul>
<li>Computer</li>
<li>Internet Facility</li>
<li>Relevant Software</li>
</ul>
</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">
<ul>
<li>Business Continuity for Dummies, The Cabinet Office, Stuart Sterling, Anna Payne, Brian Duddridge, Andrew Elliott, Michael Conway, ISBN: 9781118326831</li>
</ul>
</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">
<ul>
<li>Lectures, Presentations, Demonstration, Practical, Discussions using the material</li>
</ul>
</td></tr>
<tr><td rowspan="5">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="4">Continuous Assessment</td><td>Practical 01</td><td>15%</td></tr>
<tr><td>Practical 02</td><td>15%</td></tr>
<tr><td>Practical 03</td><td>15%</td></tr>
<tr><td>Make a presentation on BCP and DR Plans developed in practical 1 or practical 2</td><td>15%</td></tr>
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| | Summative Assessment | Written test -2 hours | 40% |
| **Duration** | **100hrs. (60T + 25P + 15S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

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|---|---|
| **Module Title** | **Generate Analytical Reports Using Data Science Tools** |
| **Module Code** | K72T001M32/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U12 |
| **Pre-Requisites** | None |
| **Module Aim/s** | To enable the students to:<br>Develop skills needed to generate analytical reports using data science tools & techniques and to teach methodologies can use for data acquisition, preprocessing, analysis, visualization, and report generation tailored for decision-making in various business contexts. |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Clean and transform large datasets using relevant tools<br>2. Manage missing values, outliers, and inconsistencies<br>3. Discover hidden patterns and relationships<br>4. Create professional-looking reports using relevant tools |
| **Learning Content / Topics** | • Data Science Tools and Techniques (12 hrs.)<br>  ○ Overview of data science and its applications<br>  ○ Key data science tools (e.g. Python, R, Excel, Tableau, Power BI, etc.)<br>• Data Acquisition and Preprocessing (12 hrs.)<br>  ○ Data sources and extraction methods<br>  ▪ APIs, databases, web scraping<br>  ○ Cleaning, wrangling, and transforming data<br>  ○ Handling missing and inconsistent data (missing values and outliers)<br>• Statistical Analysis (12 hrs.)<br>  ○ Fundamentals of statistical analysis<br>  ○ Hypothesis testing<br>• Data Visualization (06 hrs.)<br>  ○ Principles of effective data visualization<br>  ○ Creating visualizations (using Tableau, Power BI, Python libraries, etc.)<br>  ○ Designing interactive dashboards<br>• Analytical Report Generation (06 hrs.)<br>  ○ Structuring data-driven reports<br>  ▪ Executive summary<br>  ▪ Methodologies used<br>  ▪ Data analysis and summarization<br>  ▪ Findings and recommendations<br>  ○ Notification and escalation<br>• Best practices for presenting findings to stakeholders (06 hrs.) |

<table>
<tr><td>Practical</td><td colspan="3">Prepare a professional data analytical report for a given case study/scenario.<br>Report should include at least the following sections in it.<br>○ Executive summary<br>○ Methodologies used<br>○ Data analysis and summarization<br>○ Findings and recommendations</td></tr>
<tr><td>Resources</td><td colspan="3">• Computer<br>• Internet Facility<br>• Relevant Software</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Data Science for Business Professionals: A Practical Guide for Beginners, Probyto Data Science and Consulting Pvt. Ltd., ISBN: 9789389423280<br>• Data Science for Business and Decision Making, Luiz Paulo Fávero, Patrícia Belfiore, ISBN: 9780128112168</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">• Lectures<br>• Presentations<br>• Demonstration<br>• Practical<br>• Discussions using the material</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Practical</td><td>30%</td></tr>
<tr><td>Make a presentation on the generated report</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -2 hours</td><td>50%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (54T + 34P + 12S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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|---|---|
| **Module Title** | **Design, Develop and Maintain Web Application** |
| **Module Code** | K72T001M33/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U17, K72T001U18 |
| **Pre-Requisites** | |
| **Module Aim/s** | **To enable the students to:**<br>This module equips learners with the skills and knowledge required to design, develop, host, and maintain web applications. It covers various aspects of web development, including front-end and back-end development, database management, server configuration, and maintenance. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Design robust and scalable web application architectures<br>2. Develop comprehensive technical specifications for web applications<br>3. Implement sophisticated front-end and back-end functionalities<br>4. Deploy web applications effectively on various platforms<br>5. Maintain and optimize deployed web applications for performance and security<br>6. Integrate developed website with a payment gateway |
| **Learning Content / Topics** | • Web Applications Architecture - (18hrs.)<br>o Types of web applications ((e.g., static, dynamic, single-page, multiple-pages)<br>o Types of architectures (e.g. monolithic, microservices, serverless)<br>o Client-server model and its components<br>o Three-tier architecture and its layers<br>o Cloud computing and its role in web application development<br><br>• Develop Front-end - (24hrs.)<br>o HTML and CSS<br>▪ Semantic HTML5 structures.<br>▪ Graphic, Multimedia and API integration<br>▪ Responsive design using CSS frameworks<br>o JavaScript for Interactivity<br>▪ DOM manipulation<br>▪ Event handling<br>o Front-end Frameworks<br>▪ Introduction to Front-end Frameworks<br>▪ State management techniques |

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| | ○ UI/UX Principles<br>▪ UI/UX design principles<br>▪ Designing user-friendly interfaces<br>▪ Accessibility and usability best practices<br>○ Testing and Debugging<br>▪ Automated front-end testing Tools and techniques<br><br>• Payment gateways - (03hrs.)<br>○ Features (Availability, security, performance, etc.)<br>○ Integrate with web applications<br>• Develop Back-end - (18hrs.)<br>○ Server-side programming Languages<br>○ Back-end frameworks<br>○ XML<br>○ Database Integration<br>○ API Development (SOAP, JSON, etc.)<br>○ Authentication and Authorization<br>○ Error Handling and Logging<br><br>• Deploy the Web Application - (18hrs.)<br>○ Hosting Platforms (Shared hosting, VPS, cloud hosting, etc.)<br>○ CI/CD Pipelines<br>○ Version Control<br>○ Introduction to Containerization<br>○ Domain Configuration<br>○ Performance Optimization<br>○ Search Engine Optimization (SEO)<br><br>• Maintain Deployed Web Applications - (09hrs.)<br>○ Monitoring and Analytics<br>○ Bug Fixing and Updates<br>○ Backup and Recovery<br>▪ Setting up automated backups<br>▪ Disaster recovery planning<br>○ User Support |
| **Practical** | • Practical 01 - Design and develop a web application with a well-designed front end, back end, payment gateway with database integration for a given scenario (Group project)<br>• Practical 02: Prepare system documentation |

<table>
<tr><td>Resources</td><td colspan="3">• Computer<br>• Internet Facility<br>• Relevant Software</td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3">• Head First HTML and CSS, 2nd Edition Elisabeth Robson, Eric Freeman ISBN: 9780596159900<br>• Bootstrap: Responsive Web Development, 1st Edition Jake Spurlock ISBN: 9781449344597<br>• Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5 5th Edition Robin Nixon ISBN:978-1491978917<br>• JavaScript: The Definitive Guide, 7th Edition David Flanagan ISBN: 9781491952023<br>• Building Web Apps with WordPress, Second Edition, by Brian Messenlehner and Jason Coleman ISBN: 978-1-491-99008-7<br>• Full Stack Web Development the Comprehensive Guide, Philip Ackermann, ISBN: 978-1-4932-2438-8</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">• Lectures<br>• Presentations<br>• Demonstration<br>• Practical<br>• Discussions using the material</td></tr>
<tr><td rowspan="4">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="3">Continuous Assessment</td><td>Practical 01</td><td>30%</td></tr>
<tr><td>Practical 02</td><td>20%</td></tr>
<tr><td>Make a presentation for the Practical 01</td><td>10%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -2 hours</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">175hrs. (90T + 75P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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|---|---|
| **Module Title** | **Advertising and Campaign Management** |
| **Module Code** | K72T001M34/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U16, K72T001U17, K72T001U18, K72T001U19 |
| **Pre-Requisites** | K72T001M17, K72T001M22, K72T001M23 |
| **Module Aim/s** | **To enable the students to:**<br>Conduct market research to identify the market trends, consumer behaviors and design & develop advertising materials such as Scripts, Visuals, and Multimedia content to fulfil the customer requirements. |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Prepare advertising plans<br>2. Design and develop Advertising materials (Scripts, Visuals, and Multimedia content)<br>3. Manage Advertising campaigns<br>4. Manage Digital marketing campaigns |
| **Learning Content / Topics** | • Advertising concepts (15 hrs.)<br>o History and evolution of advertising.<br>o Consumer behavior and psychology.<br>o Introduction to advertising concepts and theories.<br>o Role of advertising in business and society<br>o Market research and audience analysis<br>o Evaluation of different advertising mediums<br>o Branding and brand positioning strategies.<br>o Developing advertising campaigns<br>o Ethical and legal considerations in advertising.<br>• Market Research and Analysis (12 hrs.)<br>o Conducting qualitative and quantitative research.<br>o Analyzing market trends and consumer insights.<br>o Utilizing research findings to develop advertising strategies.<br>• Media Planning and Buying (09 hrs.)<br>o Overview of different media/channels/ platforms (TV, radio, print, digital).<br>o Budgeting and scheduling media buys.<br>o Evaluating media effectiveness and return on investment (ROI).<br>• Campaign Management (09 hrs.)<br>o Planning and executing advertising campaigns. |

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| | ○ Monitoring campaign performance and making adjustments.<br><br>• Evaluation and Reporting (06 hrs.)<br>○ Key performance indicators (KPIs) use to measure campaign effectiveness.<br>○ Providing recommendations for future advertising initiatives/improvements.<br><br>• Digital Marketing Techniques (06 hrs.)<br>○ Search engine optimization (SEO) strategies<br>○ Pay-per-click (PPC) advertising<br>○ Social media marketing and content strategy<br><br>• Strategic Advertising Management (06 hrs.)<br>○ Campaign lifecycle: planning, execution, monitoring.<br>○ Budgeting and resource allocation.<br>○ Case studies of successful campaigns.<br><br>• Measuring Advertising Effectiveness (06 hrs.)<br>○ Analytics tools to measure performance of advertising campaigns<br><br>• AI in advertising (06 hrs.) |
| **Practical** | • Assignment - Prepare a report on advertising campaigns for a given scenario. The report should contain the followings:<br>○ Campaign theme<br>○ Budget details<br>○ Advertising materials (Scripts, Visuals, and Multimedia content)<br>○ Key performance indicators (KPIs)<br>○ Customer feedbacks |
| **Resources** | • Computer<br>• Internet Facility<br>• Relevant Software |
| **Prescribed Texts & / or References** | • The Advertising Concept Book: Think Now, Design Later, Pete Barry, ISBN: 9780500773604<br>• The Copywriter's Handbook: A Step-by-Step Guide to Writing Copy That Sells, 4th Edition, Robert W. Bly, ISBN: 9781250238016<br>• Advertising Media Planning, Seventh Edition, Roger Baron, Jack Sissors, ISBN: 9780071703123 |
| **Teaching/Learning Activities** | • Lectures<br>• Presentations<br>• Demonstration |

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| | • Practical<br>• Discussions using the material | | |
| **Assessment & Weighting** | **Type** | **Topic / Activity/Learning outcomes** | **Module Weighting** |
| | Continuous Assessment | Assignment | 30% |
| | | Make a presentation of prepared advertising campaigns proposal | 30% |
| | Summative Assessment | Written Exam | 40% |
| **Duration** | **125hrs. (75T + 40P + 10S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

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|---|---|
| **Module Title** | **Analyze and finalize the cost estimate of the Network solution** |
| **Module Code** | K72T001M35/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U23 |
| **Pre-Requisites** | None |
| **Module Aim/s** | **To enable the students to:**<br>• Equip learners with skills to analyze network requirements for cost estimation and to prepare, validate, and finalize network solution cost estimates and to justify cost estimates to stakeholders |
| **Learning Outcome/s** | **At the end of this module the student shall be able to:**<br>1. Analyze and estimate costs for an enterprise network solution<br>2. Develop a detailed cost proposal, including hardware, software, and services<br>**3.** Perform financial and risk analyses |
| **Learning Content / Topics** | • Cost Estimation concepts (06 hrs.)<br>o Importance of accurate cost estimation in network solutions<br>o Overview of Capital Expenditure and Operational Expenditure in network projects<br>o Factors Influencing for Costs<br>o Key challenges in cost estimation<br>o Sources of Cost Data (Market Research, Vendor Quotes, Historical Data, etc.)<br>• Analyzing Network Requirements (06 hrs.)<br>o Gathering technical and business requirements<br>o Identifying scope and constraints<br>o Developing requirement specifications<br>• Cost Components in Network Solutions (09 hrs.)<br>o Hardware Cost (Servers, routers, switches, cables, accessories, etc.)<br>o Software Cost (Licensing, subscription, updates, replacements, etc.)<br>o Operational Cost (Power, internet, staff salaries, etc.)<br>o Maintenance and Support Costs<br>o Contingency Planning and Reserves Costs<br>o Other Costs (Depreciation, Commissions, etc.)<br>o Tax and regulatory considerations in cost estimation<br>o Cost justification (Net Present Value (NPV) and Discounted Cash Flow (DCF))<br>o Comparative analysis (Build vs. lease options)<br>• Legal and Compliance Considerations (03 hrs.)<br>o Compliance with procurement policies |

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| | ○ Understanding taxes, duties, and import/export regulations<br>○ Intellectual property and software licensing costs<br><br>• Risk Management in Cost Estimation (03 hrs.)<br>○ Identifying and assessing risks in network projects<br>○ Mitigation strategies for cost-related risks<br><br>• Tools and Techniques for Cost Estimation (12 hrs.)<br>○ Cost estimation tools (Spreadsheet, Specialized software)<br>○ Total Cost of Ownership (TCO), Break Even Point (BEP), Internal Rate of Return (IRR), Pay Back Period (PBP), and Return On Investment (ROI) calculations<br>○ Vendor comparison techniques<br><br>• Cost Optimization Techniques (03 hrs.)<br>○ Leveraging bulk purchasing and volume discounts<br>○ Open-source vs. proprietary solutions<br>○ Energy-efficient networking and green IT practices<br><br>• Budgeting and Financial Management (09 hrs.)<br>○ Types of costs (fixed, variable, direct, indirect)<br>○ Types of budgets (fixed, flexible, zero-based)<br>○ Difference between budget and cost estimation<br>○ Cost-benefit analysis<br>○ Developing Project Budgets<br>○ Tracking Project Expenses<br>○ Cost Control and Variance Analysis<br>○ Financial Reporting<br><br>• Validating and Finalizing Cost Estimates (06 hrs.)<br>○ Reviewing cost estimates for accuracy<br>○ Collaboration with procurement teams<br>○ Aligning cost estimates with organizational budgets |
| **Practical** | • Prepare a cost estimation report for a given enterprise network solution (may be a hypothetical network project). The report should include the followings:<br>○ Detailed cost estimate for each and every component that will be used for solution<br>○ Vendor comparison<br>○ Additional cost component (such as contingencies) |

<table>
<tr><td>Resources</td><td colspan="3"><ul><li>Multimedia, Whiteboard</li><li>Smart board</li><li>Computer lab with internet facility</li><li>Cost Estimation tools (e.g. Spreadsheet, etc.)</li><li>Cost Estimation templates</li></ul></td></tr>
<tr><td>Prescribed Texts & / or References</td><td colspan="3"><ul><li>The Art of Network Architecture, Russ White, Denise Donohue, [No ISBN]</li><li>Cost Estimation: Methods and Tools, Gregory K. Mislick, Daniel A. Nussbaum, ISBN: 9781118536131</li></ul></td></tr>
<tr><td>Teaching/ Learning Activities</td><td colspan="3"><ul><li>Lectures</li><li>Presentations</li><li>Discussions using the material and product samples</li></ul></td></tr>
<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="2">Continuous Assessment</td><td>Practical</td><td>40%</td></tr>
<tr><td>Make a presentation on the prepared cost estimation report in the practical to a panel of stakeholders. Learner should justify the investment of the project</td><td>20%</td></tr>
<tr><td>Summative Assessment</td><td>Written Exam -2 hour</td><td>40%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (57T + 30P + 13S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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|---|---|
| **Module Title** | **Fundamentals of Information, Communication & Network Security (Governance Risk and Compliance)** |
| **Module Code** | K72T001M36/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U24 |
| **Pre-Requisites** | K72T001M30, K72T001M31, K72T001M32, K72T001M33 |
| **Module Aim/s** | To enable the students to:<br>Equip learners with a comprehensive understanding of information, communication, and network security principles, with a strong emphasis on governance, risk, and compliance (GRC) frameworks |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Explain the principles of cybersecurity, governance, risk, and compliance<br>2. Identify and mitigate risks in information and network security<br>3. Apply governance frameworks to ensure robust security policies<br>4. Ensure compliance with legal and regulatory standards in cybersecurity |
| **Learning Content / Topics** | • Introduction to Cybersecurity (12 hrs.)<br>o Cybersecurity Basics (Concepts, Goals, and Challenges)<br>o CIA Triad (Confidentiality, Integrity, and Availability)<br>o Types of Cyber Threats (Malware, Phishing, Ransomware, Insider Threats, etc.)<br>o Overview of Security Domains (Physical, Network, Application, and Data Security, etc.)<br>• Information Security Governance (12 hrs.)<br>o Governance Principles and Best Practices<br>o Security Policy Framework<br>o Security Roles and Responsibilities in an Organization<br>o Aligning IT Security with Business Objectives<br>• Risk Management (06 hrs.)<br>o Understanding Risk (Identification, Assessment, and Mitigation)<br>o Risk Management Frameworks<br>o Threat Modeling and Vulnerability Assessment<br>o Developing and Implementing Risk Mitigation Plans<br>• Network Security Concepts (06 hrs.)<br>o Network Architecture and Protocols<br>o Firewalls, Intrusion Detection and Prevention Systems (IDPS)<br>o Virtual Private Networks (VPNs) and Secure Communication |

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| | ○ Network Security Best Practices<br><br>• Communication Security (06 hrs.)<br>○ Securing Email and Messaging Platforms<br>○ Encryption Techniques (Symmetric, Asymmetric, Hashing)<br>○ PKI and Digital Certificates<br><br>• Incident Response and Business Continuity (06 hrs.)<br>○ Incident Response Planning and Execution<br>○ Security Information and Event Management (SIEM)<br>○ Security operation Center (SoC)<br>○ Business Continuity and Disaster Recovery Planning<br>○ Role of Governance in Incident Management<br>○ Post-Incident Review and Reporting<br><br>• Emerging Trends and Challenges (12 hrs.)<br>○ Cloud Security Governance<br>○ Securing IoT and Mobile Devices<br>○ AI and Machine Learning in Risk Management and cybersecurity<br>○ Cybersecurity Trends and Their Impact |
| **Practical** | • Group Project 1: Develop a Governance and Risk Management Framework (for a small enterprise)<br>• Group Project 2: Develop an information security policy (for a small enterprise)<br>• Group Project 3:<br>Conduct a Security Awareness and Training Session<br>○ Develop Security Awareness Training Materials: Create presentations, videos, or other materials to educate employees about security risks and best practices<br>○ Conduct Security Awareness Training Sessions: Deliver training sessions to employees on topics such as phishing, social engineering, and safe password practices<br>○ Evaluate the Effectiveness of Security Awareness Training: Measure the impact of training through surveys, quizzes, or simulated phishing attacks<br>• Group project 4: Prepare vulnerability assessment report for a selected company |
| **Resources** | • Computer<br>• Internet Facility<br>• Relevant Software |
| **Prescribed Texts & / or References** | • Computer and Information Security Handbook - 3rd Edition, John R. Vacca<br>• The Complete Cyber Security Course, Volume 1 Hackers Exposed, Nathan |

<table>
<tr><td></td><td colspan="3">House<br>• Principles of Information Security - 5th Edition, Whitman, Michael Mattord, Herbert<br>• Foundations of Information Security A Straightforward Introduction, Jason Andress<br>• Cybersecurity for Dummies, Joseph Steinberg<br>• Network Security for Dummies, 1st Edition, Chey Cobb</td></tr>
<tr><td>Teaching/Learning Activities</td><td colspan="3">• Lectures<br>• Presentations<br>• Demonstration<br>• Practical<br>• Discussions using the material</td></tr>
<tr><td rowspan="6">Assessment & Weighting</td><td>Type</td><td>Topic / Activity/Learning outcomes</td><td>Module Weighting</td></tr>
<tr><td rowspan="4">Continuous Assessment</td><td>Group Project 01</td><td>15%</td></tr>
<tr><td>Group Project 02</td><td>15%</td></tr>
<tr><td>Group Project 03</td><td>15%</td></tr>
<tr><td>Group Project 04</td><td>15%</td></tr>
<tr><td>Summative Assessment</td><td>Make a presentation based on one of the selected above Projects<br>Written Exam -2 hours</td><td>10%<br>30%</td></tr>
<tr><td>Duration</td><td colspan="3">100hrs. (60T + 30P + 10S)<br>(T-Theory, P-Practical, S- Self Studies)</td></tr>
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| | |
|---|---|
| **Module Title** | **Ethical Hacking, Mobile and Internet of Things (IoT) Security** |
| **Module Code** | K72T001M37/ Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | K72T001U25, K72T001U27 |
| **Pre-Requisites** | K72T001M30, K72T001M31 |
| **Module Aim/s** | To enable the students to:<br>Equip students with essential knowledge in ethical hacking principles and techniques, secure mobile platforms, and Internet of Things (IoT) devices, focusing on identifying vulnerabilities and mitigating threats. |
| **Learning Outcome/s** | At the end of this module the student shall be able to:<br>1. Gain an understanding of ethical hacking techniques, methodologies, tools and frameworks<br>2. Learn to identify and mitigate security risks in mobile devices and applications<br>3. Identify and address security challenges specific to IoT devices and networks<br>4. Apply best practices for securing mobile and IoT ecosystems<br>5. Adhere to industry best practices and ethical guidelines |
| **Learning Content / Topics** | • Ethical Hacking (12 hrs.)<br>o Ethical hacking concepts and principles<br>o Ethical hacking methodologies (e.g., penetration testing, vulnerability scanning)<br>o Legal and ethical considerations in ethical hacking<br>o Foot printing and reconnaissance techniques<br>o Vulnerability scanning and exploitation tools<br>• Mobile Device Security (6 hrs.)<br>o Mobile device security threats and vulnerabilities<br>o Mobile device operating systems and security features<br>o Mobile application security<br>o Mobile device forensics<br>o Mobile device management (MDM) solutions<br>• Internet of Things (IoT) Security (12 hrs.)<br>o IoT devices and their security challenges<br>o IoT protocols and vulnerabilities<br>o IoT security standards and best practices<br>o Securing IoT networks and devices<br>o IoT threat intelligence and incident response |

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| **Practical** | • Practical 1: Vulnerability Scanning -Use tools like Nmap, Nessus, or OpenVAS to scan a target system (e.g., a virtual machine) for vulnerabilities. Analyze the scan results and identify potential risks.<br>• Practical 2: Penetration Testing -Conduct a simulated attack on a target system (e.g., a web application) to identify and exploit vulnerabilities. This could involve techniques like SQL injection, cross-site scripting (XSS), and buffer overflows.<br>• Practical 3: Social Engineering -Conduct simulated phishing attacks or social engineering exercises to assess the awareness and resilience of individuals and organizations.<br>• Practical 4: Mobile Device Security -Analyze and configure mobile device security settings, such as passcodes, passwords, encryption, and app permissions.<br>• Practical 5: Mobile Application Security -Analyze mobile applications for vulnerabilities, such as insecure data storage, insecure communication, and privacy violations.<br>• Practical 6: Mobile Malware Analysis -Analyze suspicious mobile applications using sandboxing and other techniques to detect malware. | | |
| **Resources** | • Computer<br>• Internet Facility<br>• Relevant Software | | |
| **Prescribed Texts & / or References** | • Computer and Information Security Handbook - 3rd Edition, John R. Vacca<br>• The Complete Cyber Security Course, Volume 1 Hackers Exposed, Nathan House<br>• Principles of Information Security - 5th Edition, Whitman, Michael Mattord, Herbert<br>• Foundations of Information Security A Straightforward Introduction, Jason Andress<br>• Cybersecurity for Dummies, Joseph Steinberg<br>• Network Security For Dummies, 1st Edition, Chey Cobb<br>• CEH v10 Certified Ethical Hacker Study Guide, by Ric Messier<br>• CEH Certified Ethical Hacker All-in-One Exam Guide, Author Matthew Walker | | |
| **Teaching/Learning Activities** | • Lectures<br>• Presentations<br>• Demonstration<br>• Practical<br>• Discussions using the material | | |
| | **Type** | **Topic / Activity/Learning** | **Module Weighting** |

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| **Assessment & Weighting** | | **outcomes** | |
| | Continuous Assessment | Make two presentations based on two of the above practical | 30% |
| | | | 30% |
| | Summative Assessment | Written test -2 hours | 40% |
| **Duration** | **100hrs. (60T + 30P + 10S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

# MODULE DESCRIPTORS
# NVQ Level 06
# Information and Communication Technology
## Semester 2
## Employability modules

| | |
|---|---|
| **Module Title** | **Teamwork and Leadership** |
| **Module Code** | EMTM05/Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | EMTU05 |
| **Pre-Requisites** | None |
| **Module Aims** | **To enable the students to**<br>▪ Acquire the skills of working with others as a team<br>▪ Understand the need of maintaining positive relationships with others under varying work conditions.<br>▪ Lead and facilitate work teams to achieve planned outcome while enhancing organizational productivity |
| **Learning Outcomes** | **The student shall be able to**<br>▪ Define objectives and functions of teams to develop team plans towards achieving corporate goals of the organization<br>▪ Identify the important aspects of motivating individuals/teams and enhancing co-operation and commitment<br>▪ Demonstrate an understanding of analyzing ideas and information through discussion and making informed decisions.<br>▪ Manage and develop team performance by applying appropriate techniques and methods<br>▪ Apply understanding of roles of team leadership and membership to build trust and confidence<br>▪ Encourage members in the team to take initiative and make innovations |
| **Learning Content / Topics** | • Introduction<br>• Functions and roles of leadership, teams in an organization<br>• Leadership styles and Leadership roles<br>• Team processes, identifying your role within a team, How a team develops, Team planning- defining objectives, purpose functions and accountabilities<br>• Discussion techniques<br>• Communication in a team environment<br>• Decision making techniques<br>• Leadership and personality development<br>• Social analysis - culture and values, psychological and social aspects of individuals |
| **Resources** | **Occupational Tools, Instruments, Equipment, material**<br>• Documents on Company policies and procedures<br>• Calculator |

<table>
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- Computer
- Documents on Industry / Workplace Codes of Practice/ Codes of ethics
- Documents on Manufacturer' s specifications and product specifications
- Safety signs
- Safety procedures
- Variety of forms, formats used within the organization

**Instructional Tools, Instruments, Equipment, material**

- Multimedia projector
- Screen
- Flash cards
- Flip charts
- Permanent and white board markers
- Overhead projector
- Transparencies
- Color Printer and scanner
- Safety manuals
- Safety illustrations

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<tr><td>Recommended Texts & / or References</td><td colspan="3"></td></tr>
<tr><td>Teaching Learning Activities</td><td colspan="3">

- Lectures for imparting the fundamental aspects teamwork and leadership
- Brainstorm appropriate interpersonal skills for working with and for others
- Conduct exercises in team planning
- List out in small groups attitudes of workers that create a positive working atmosphere
- Brainstorm how to develop commitment and cooperation within a team
- How to manage and develop team performance
- Prepare a check list that would outline the main steps in developing team performance

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<tr><td rowspan="3">Assessment & Weighting</td><td>Type</td><td>Topic / Activity</td><td>Weighting</td></tr>
<tr><td rowspan="2">Formative Assessment</td><td>Skills in application of Leading and facilitating work teams at workplace during industry training</td><td>50%</td></tr>
<tr><td>knowledge on Leading and facilitating work teams -</td><td>10%</td></tr>
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| | | Oral questioning technique during class room presentations | |
| | Summative Assessment | knowledge on Leading and facilitating work teams - Multiple Choice Test Items/ Matching and Completion test items and structured essay type questions | 30% |
| | | Viva voce- knowledge in application of Leading and facilitating work teams | 10% |
| **Duration** | 50 hours **(30T + 15P + 05S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

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|---|---|
| **Module Title** | **Creating & maintaining a learning culture at workplace** |
| **Module Code** | EMTM06/Level 06 |
| **Module Type** | Compulsory |
| **Relevant Unit/s** | EMTU06 |
| **Pre-Requisites** | None |
| **Module Aims** | **To enable the students to**<br>▪ Create awareness and understanding of the concept of learning culture.<br>▪ Develop competencies required to create and maintain a learning culture at the workplace. |
| **Learning Outcomes** | The student shall be able to<br>▪ Inform management about training requirements of workers<br>▪ Arrange /Conduct training programs<br>▪ Demonstrate the use of new equipment and methods |
| **Learning Content / Topics** | ▪ Definition of training, definition of performance, performance appraisal, methods of identifying performance gaps of workers, how to design training programs<br>▪ Setting of learning outcomes, motivating workers for participation at training programs, Hints and tips on conducting demonstrations<br>▪ Practice - Arrange /Conduct training programs<br>▪ Demonstrate the use of new equipment and methods |
| **Practical** | |
| **Resources** | • Hand outs<br>• Computer<br>• Printer<br>• Photocopier<br>• Multimedia projector<br>• Models<br>• Transparencies<br>• Manuals |
| **Prescribed Texts & / or References** | |
| **Teaching Learning Activities** | ▪ Use lectures to impart theoretical knowledge on various aspects of learning culture.<br>▪ Use group discussions, case studies, Role plays, to develop required skills and attitudinal aspects. |

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| | ▪ Create and maintain a positive learning culture at the training center.<br>▪ Arrange guest lectures on learning culture. | | |
| **Assessment & Weighting** | **Type** | **Topic / Activity** | **Weighting** |
| | Formative Assessment | skills and attitudes related to various aspects of learning culture at work place - Role plays and simulations | 40% |
| | Summative Assessment | Knowledge of various aspects of learning culture at work place -Multiple Choice Test Items, structured essay type questions, case studies | 60% |
| **Duration** | 50 hours **(30T + 15P + 05S)**<br>**(T-Theory, P-Practical, S- Self Studies)** | | |

## Competency Based Training Curriculum
## For NVQ Levels 5 National Diplomas in Information and Communication Technology

| **Policy Direction** |
|---|
| • Mr. Tharanga Naleen Gamlath, Chairman, National Apprentice & Industrial Training Authority<br>• Prof. Mahinda Rupasinghe, Vice Chairman, National Apprentice & Industrial Training Authority<br>• Dr. K.A. Lalithadeera, Director General, Tertiary & Vocational Education Commission<br>• Ms. Padma Samaradiwakara, Directress (Quality), National Apprentice & Industrial Training Authority |
| **Revision Group** |
| • Mr.Lalith Wasantha –Head of Engineering STEIN Studios<br>• Mr.Nadith Kudagama-Chief -Technical Officer/Eyepax Pvt Ltd<br>• Mr. Helawikum Atauda Seneviratne-Dean ICT-UNIVOTEC/Senior Lecturer<br>• Mrs. N. M. P. Gunawardana-Lecturer ICT (SLTES) –Sri Lanka College of Technology-Maradana<br>• Mr. M. K. G. Perera-Lecturer ICT(SLTES)-Sri Lanka College of Technology- Maradana<br>• Mr. K.T.C. Devaka De Silva- Lecturer ICT(SLTES)-Sri Lanka College of Technology- Galle<br>• Dr. Damitha Karunarathna- University of Colombo<br>• Mr. Sudharshan Welihinda- Head of Software Development /Business unit at InformaticsInternational Limited.<br>• Mr. Ashan De Soyza-Project Director - LAS ICT Management Consultants Pvt Ltd |
| **National Industrial Training Advisory Committee** |
| • Dr. Viraj Pinto, Chairman – Matrix Institute of Information Technology (Pvt.) Ltd., Matrix totalenterprise solutions (Pvt.) Ltd., Matrix consultancy Service and Solutions (Pvt.) Ltd.<br>• Mr. Sellam Ambalavanar, Former DGM – Call Center Technical Supports, Sri Lanka Telecom<br>• Mr. W.C. Priyashantha, Former Project Lecturer, School of computing, University of Colombo<br>• Mr. Sameera Jayawardena, Director- Capacity Building – ICTA<br>• Mr. Thilina Dissanayake, Manager - Capacity Building – ICTA<br>• Mr. Damian Fernando, Founder / Managing Director, Paperrockets Pvt. Ltd.<br>• Mr. Kapila Giragama, Manager – ICT Sector Skills Council<br>• Dr. Shanta Yapa, Member - ICT Sector Skills Council<br>• Dr. Dilrukshi Herath, Assistant Director – TVEC |
| **Facilitators- Revision** |
| • Mr. Nihal Dias-Former Director-NAITA<br>• Mr. Helawikum Atauda Senevirathna-Dean ICT-UNIVOTEC/Senior Lecturer |
| **Coordinator- Revision** |
| • Mr. G.D.I. Pushpakumara, Associate Officer, University of Vocational Technology |
| **Facilitators - Validation** |

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|---|
| • Mrs. M.P. Theja Malkanthi, Assistant Director (SSCD & LRD), National apprentice & Industrial Training Authority, Rajagiriya |
| **Coordinator- Validation** |
| • Ms. W.A.S.M. De Silva<br>Inspector (SSCD), National apprentice & Industrial Training Authority, Rajagiriya |